Giving Shells a New Life: Oyster Waste Solutions in Ang Sila

ชุบชีวิตให้เปลือกหอย: โซลูชันการจัดการขยะเปลือกหอยนางรมในอ่างศิลา

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Giving Shells a New Life: Oyster Waste Solutions in Angsila

ชุบชีวิตให้เปลือกหอย

Developing Solutions for Oyster Shell Waste Management and Repurposing

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Abstract

Oyster farming is central to Angsila's economy, but the increasing volume of oyster shell waste has created management challenges. Shells are left in open piles, causing odor, pests, and environmental health risks. However, oyster shells can be a valuable resource with potential in agriculture and construction. This project proposes a viable waste management and repurposing system by identifying gaps in current practices, developing a scalable processing model for value-added products, and evaluating market feasibility. Using field observations, interviews, material testing, and cost analysis, the study found that cleaning can reduce environmental impact and enable shells to be converted into raw materials, with recommendations for phased implementation from low-tech to industrial systems.

บทคัดย่อ การเพาะเลี้ยงหอยนางรมเป็นกิจกรรมที่มีความสำคัญต่อเศรษฐกิจของอ่างศิลาเป็นอย่างมาก แต่ปริมาณเปลือกหอยนางรมที่เพิ่มขึ้นอย่างต่อเนื่องก่อให้เกิดปัญหาในการจัดการของเสีย โดยเปลือกหอยมักถูกกองทิ้งไว้ในพื้นที่โล่งกลางแจ้ง ทำให้เกิดกลิ่นไม่พึงประสงค์ ซึ่งกลายเป็นแหล่งเพาะพันธุ์ของแมลง นำมาซึ่งความเสี่ยงต่อสุขภาพและสิ่งแวดล้อมในบริเวณใกล้เคียงแหล่งชุมชน อย่างไรก็ตาม เปลือกหอยนางรมเป็นทรัพยากรที่มีมูลค่า และมีศักยภาพในการนำไปใช้ประโยชน์ในด้านต่างๆ ทั้งด้านการเกษตรและอุตสาหกรรมก่อสร้าง โดยโครงการนี้มีวัตถุประสงค์ที่จะเสนอแนวทางการจัดการขยะเปลือกหอยกลับมาใช้ประโยชน์อย่างเหมาะสม โดยการระบุช่องว่างในระบบการจัดการปัจจุบัน พัฒนารูปแบบกระบวนการแปรรูปที่สามารถขยายกำลังการผลิตได้ ประเมินความเป็นไปได้ทางการตลาดและการขยายผล การศึกษานี้ใช้การลงพื้นที่สังเกตการณ์ การสัมภาษณ์ การทดสอบวัสดุ และการวิเคราะห์ต้นทุน ผลการศึกษาพบว่าการรวบรวมและแปรรูปแบบศูนย์กลางร่วมกับกระบวนการทำความสะอาดที่มีประสิทธิภาพและสามารถลดผลกระทบด้านสิ่งแวดล้อมและเปลี่ยนเปลือกหอยให้เป็นวัตถุดิบที่ใช้ประโยชน์ได้พร้อมข้อเสนอแนะในการดำเนินการแบบเป็นระยะเบื้องต้นจากการผลิตในสเกลขนาดเล็กและระบบการผลิตในสเกลขนาดใหญ่ในระดับอุตสาหกรรม

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Executive Summary

0.1 Establishing the Core Oyster Shell Waste Problem

For generations, oyster farming has sustained Angsila, Thailand's economy and livelihoods. The coastal community of Angsila is located in one of Thailand's largest oyster production areas, with a mixture of farms that use semi-traditional and cultural techniques (Explore Thailand Through Photography, n.d.). However, the increasing demand for oysters means that about seven tonnes of shell waste are discarded daily along beach coasts, roads, and nearby residential housing, where residual oyster meat decays and produces gaseous pollutants such as ammonia and hydrogen sulfide. These conditions create a breeding ground for bacteria and pests that directly affect surrounding communities. With no proper system for cleaning and managing these shells, the easiest and most common method of disposal is by dumping them illegally on coastal areas. Many locals are unaware of the dangers that this poses and are, in turn, unintentionally fueling the rising health risks in the community.

0.2 Our Goal

Our project seeks to positively impact the municipality of Angsila and Chonburi Province by co-developing an effective and feasible cleaning and sanitizing process to convert waste into beneficial products for the community. The APEC "Guidelines on the Reduction of Seafood Waste and Loss" recognize shell waste accumulation as a challenge, noting it results in "direct effects on the environment" and calling for value-added reuse instead (C. Godoy, Asia-Pacific Economic Cooperation, 2022).

0.3 Methodology

The goal of this project was to develop various solutions for repurposing oyster shell waste in the Angsila Municipality. To achieve this goal, the team developed and executed a methodology structured around three objectives: identifying gaps and capacities in the existing waste management system, developing a practical method for processing oyster shells into usable materials, and analyzing the market feasibility and scalability of oyster shell repurposing.

To support these objectives, the team employed a combination of stakeholder interviews, site observations, material sampling, process design, and market and cost analysis. Data collected through these methods informed the development of recommendations for a waste-effective and community-appropriate shell management system.

0.3.1 1. Identifying gaps and potential capacities in the current waste management system

To understand current oyster shell disposal practices, the team consulted with Angsila Town Municipality officials and conducted site visits to multiple oyster shucking facilities.

According to the municipality, small shucking houses on average produce 2-3 tonnes of oyster shell waste per day, while large shucking houses on average produce 4-5 tonnes per day.

Findings from interviews and observations indicated that dispersed dumping sites are difficult to monitor and manage, contribute to persistent odor issues, and create conditions favorable for bacterial growth and pests. Although workers did not identify immediate health concerns, the long-term environmental and public health risks were evident. Nylon rope and concrete residue from current oyster farming methods were also identified as major barriers to reuse. These findings highlighted the need for centralized disposal, improved cleaning methods, and increased awareness of the potential value of oyster shells.

0.3.2 2. Develop a scalable model for how shells should be processed into raw materials for diverse value-added applications

Based on stakeholder input and field observations, the team developed a proposed processing workflow centered on relocating shell waste away from residential areas and preparing shells for reuse. Meetings with municipal officials and community stakeholders allowed the team to align proposed solutions with local priorities and constraints.

The proposed system begins with the establishment of a centralized disposal site capable of accommodating the municipality's estimated weekly shell waste volume. Centralized collection would reduce odor and environmental impacts while enabling consistent processing and oversight. Municipal interviews revealed consistent concerns regarding strong odors due to decomposing residual flesh, fly infestations, and the difficulty of managing large volumes of shell waste. While workers expressed limited concern about long-term environmental or health impacts, odor was cited as the most immediate and disruptive issue. With the implementation of a shared disposal area, shucking houses would be encouraged to contribute shells to this system, rather than relying on informal waste disposers.

To address the odor complaints from the locals, the team explored the use of Calcium Hydroxide, Effective Microorganisms, and Photosynthetic Bacteria as a spray-on solution to reduce odor.

To address contamination from farming materials, the team proposed and pilot-tested a simple separation method in which shells were roughly crushed to separate the shell and concrete fragments away from the nylon rope, allowing the rope to be removed, followed by sieving to separate the concrete fragments. After separation, crushed shells underwent cleaning through water-bath agitation to remove residual organic matter. After cleaning, shells were air-dried for several days in well-ventilated or outdoor areas to further sanitize the material. Once dried, the shells were ground into a fine particulate that can be easily marketed for downstream use.

Our proposed and tested shell repurposing prototypes included pavers and tiles/coasters. Pavers were produced by mixing coarse ground oyster shells with cement and compressing the mixture in molds using a hydraulic press, allowing larger shell fragments to be utilized while achieving sufficient strength. Coasters were produced using oyster shell powder mixed with cement and water, which was poured into molds and compressed with a hydraulic press to increase density and mechanical strength. This process is similar to tile production and can be

adapted for future tile manufacturing by modifying the mold design.

0.3.3 3. Analyzing market feasibility and future scalability

To evaluate whether oyster shell repurposing could be economically viable and scalable, the team conducted a market analysis using both primary and secondary data sources.

To assess processing feasibility at different scales, the team analyzed two operating scenarios: a small-scale pilot model using low-tech equipment and an industrial model designed for higher throughput. Market analysis focused on business-to-business for pavers and tiles, and business-to-consumer market for coasters.

0.4 Results

We found that oyster shell waste in Angsila can be managed more effectively through coordinated collection, odor reduction, and repurposing strategies that align with local capacities and market demand. The primary barriers to effective shell waste management are not technical limitations, but rather fragmented disposal practices, odor concerns, and the absence of a structured system to convert shells into usable products.

0.5 Environmental and Health Implications

Observations and stakeholder input indicate that current disposal practices exacerbate environmental degradation and community nuisance. The presence of nylon rope and concrete from farming practices further complicates reuse and contributes to plastic pollution risks.

0.6 Odor Reduction Testing

Our experiments showed that biological treatment methods may be able to reduce foul odors associated with oyster shells by removing residual organic matter. Combinations of photosynthetic bacteria, effective microorganisms, and calcium hydroxide show potential as solutions that can be sprayed over shell piles to mitigate odor. Participant-based odor ranking tests were inherently subjective, reflecting individual differences in human odor perception and personal sensitivity. Environmental testing further indicated that some treatments produced alkaline conditions, highlighting the need for additional testing to refine odor control methods and ensure minimal environmental impact before large-scale implementation.

0.7 Material Testing and Repurposing Feasibility

Material testing demonstrated that crushed oyster shell-concrete mixtures exhibit properties suitable for selected non-structural applications. Water absorption testing showed increased porosity with higher shell content, indicating suitability for products such as pavers that benefit from permeability. Flexural strength testing revealed a predictable decrease in strength as oyster shell content increased, allowing product applications to be matched to appropriate shell ratios.

Based on these results, the most feasible repurposing strategy is to process oyster shells into raw shell powder, which can then supply multiple downstream applications. Shell powder demonstrated broad applicability across construction, agriculture, and consumer products, reducing reliance on a single market and increasing system resilience.

0.8 Economic and Market Analysis

Cost and market analyses indicate that oyster shell repurposing is economically viable under both small-scale and industrial-scale production scenarios. For construction products such as pavers and tiles, oyster shell–based materials are cost-competitive with conventional alternatives while offering added environmental value. For consumer products such as coasters, higher-margin B2C markets—particularly tourism-driven markets—offer strong revenue potential when paired with sustainability storytelling.

0.9 Recommendations

Our findings informed a phased implementation strategy in which Angsila first prioritizes improved collection and small-scale processing of oyster shells into oyster shell powder following the recommended protocol, followed by expansion into industrial-scale processing and diversified markets as capacity and demand increase.

Based on different repurposing methods, such as tiles, pavers, and coasters, different business approaches are needed to ascertain different audiences and markets. For marketing tiles and pavers, it is recommended that businesses have a B2B approach, while coasters have a B2C approach (Pilelienė, 2019). Overall, the results support a phased implementation approach in which Angsila prioritizes improved collection and basic processing, followed by expansion into diversified markets as capacity and demand increase.

0.9.1 Future Capabilites

- Another method for repurposing oyster shell waste is breakwaters and shorelines, which have been implemented in other countries and can utilize whole shells, removing the need to crush them.
- While microbial odor-reduction approaches may be feasible, further refinement of the process is needed to establish statistically conclusive results. Future work would include optimizing the treatments to improve odor mitigation on a larger scale.
- An alternative method for cleaning the shells is with the use of biosurfactants. Biosurfactants are biodegradable compounds produced by microorganisms that reduce surface tension to break down residual organic matter. This cleaning method would allow for safe repurposing of oyster shells for cosmetic purposes and allow products with higher margins to be created.

0.10 Conclusions

This project establishes that with the proper systems put in place, oyster shells can be transitioned from perceived waste into valuable and sustainable resources that generate economic opportunities. By combining environmental objectives with economic benefits, the system has the potential to reduce ecological impacts, enhance community well-being, and contribute to Thailand's circular economy. Furthermore, the insights gained from this project could guide other coastal communities facing similar challenges, making Angsila a leading example of sustainable coastal resource management.

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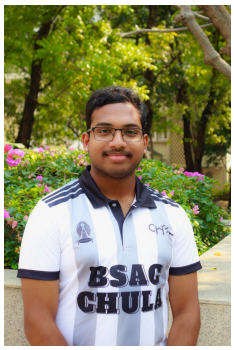

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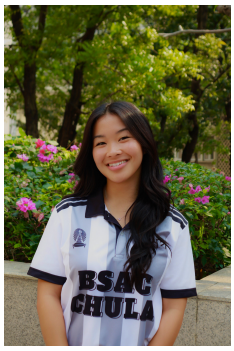

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1 Introduction

Giving Shells a New Life

ชุบชีวิตให้เปลือกหอย

Oyster shell waste has emerged as a significant environmental and management problem in many coastal and fisheries-based communities in Southeast Asia (Chilakala et al., 2019). Rapid growth in aquaculture and seafood consumption means that large volumes of oysters are harvested, shucked, and processed every day, leaving behind shells mixed with flesh residues, ropes, and other debris (Figure 1). When these shells are piled in open areas, dumped along shorelines, or sent untreated to landfills, they can cause serious odor problems and attract insects and pests, which contribute to local pollution that affects both ecosystems and public health (Siddiqua et al., 2022). At the same time, this unmanaged shell waste represents a loss of economic opportunity for communities that rely heavily on marine resources for their livelihoods.

Figure 1: Oyster Shell Waste Piles in Angsila, Thailand

Among coastal by-products, discarded oyster shells are one of the most abundant yet least utilized resources (Metalocus, 2023). Global aquaculture and capture fisheries produce millions of tonnes of oyster shells each year, and in many regions, the dominant practice is still simple disposal rather than recovery or recycling. Oyster shells are primarily composed of calcium carbonate and have demonstrated potential as raw materials for construction composites,

fertilizer, and other value-added applications, yet they remain an untapped resource (Bellei et al., 2023). The core management problem, therefore, is not only the volume of shells produced but also the lack of organized systems to collect, clean, process, and channel these shells into productive uses.

Many countries have implemented policies that follow the 3Rs (reduce, reuse, recycle) to address concerns regarding resource efficiency and waste. These policies have been implemented in many circular economy strategies, such as utilizing waste and recycling materials to maximize value (Mohan, Mohanty, Cross, & Khajuria, 2023). However, oyster shell waste is rarely addressed as a distinct stream within these frameworks. In Thailand, national initiatives linked to the Bio-Circular-Green (BCG) Economy Model push for increased recycling rates and encourage community participation, but they focus largely on municipal solid waste, such as plastics, household garbage, and organic waste (BCG, 2021). As a result, sector-specific by-products like oyster shells and aquaculture residues often fall into a policy and implementation gap. Oyster farms produce oyster shell waste in quantities, and the waste creates clear problems at the local level, but are not supported by dedicated guidelines, infrastructure, or incentive mechanisms for sustainable management. Therefore, addressing the oyster shell waste problem requires targeted approaches that recognize shells as both an environmental burden and a potential resource within the circular economy.

Angsila, a densely populated coastal municipality in Chonburi Province, Thailand, is Thailand's second-largest oyster production area with sixteen shucking houses. Every day, oysters are harvested, shucked, and processed for sale to tourists and residents. Currently, most shells are illegally discarded in open areas along beach coasts, roads, and nearby residential housing with leftover meat and nylon ropes employed from the farming process (Figure 2). An average of thirty five tonnes of waste accumulate every week from all shucking houses in the municipality, where the residual oyster meat decays and produces gaseous pollutants such as ammonia and hydrogen sulfide, with unpleasant odors.

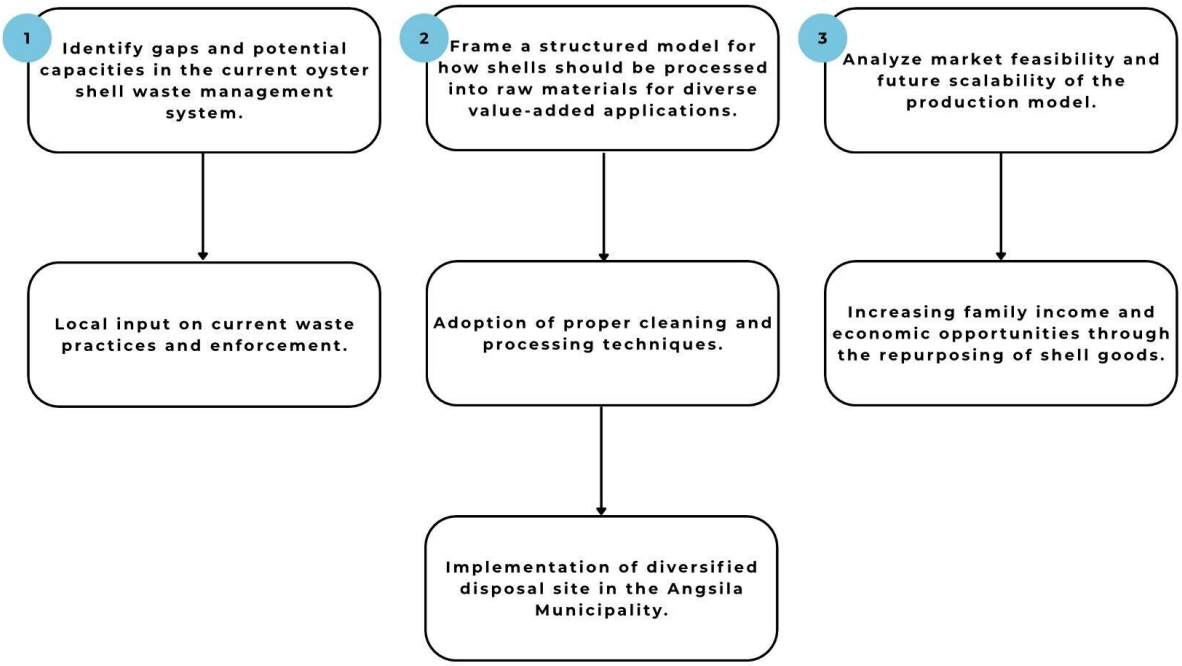
Figure 2: Oyster Shells After Shucking

These conditions reveal a critical gap between national-level waste management aspirations and the reality experienced by communities in Angsila, despite the high potential of oyster shells as a local resource for value-added products such as eco-friendly building materials or community-based handicrafts. There is a need for a more refined and structured model that guides how shells should be collected, cleaned, stored, and distributed for further processing. Moreover, local perspectives and needs regarding shell waste management have not been thoroughly investigated and documented. Without understanding the experiences of oyster farmers, market vendors, residents, and municipal officers, it is challenging to design interventions that are technically feasible, socially acceptable, economically viable, and up to the task of managing this magnitude of waste.

1. Identify gaps and potential capacities in the current oyster shell waste management system.
2. Develop a scalable model for how shells should be processed into raw materials for diverse value-added applications.
3. Analyze market feasibility and future scalability of the production model.

Our project sought to positively impact the municipality of Angsila and Chonburi Province by co-developing an effective and feasible cleaning and sanitizing process to convert waste into beneficial products for the community. By introducing repurposing methods that could be carried out locally, we aimed to create skill-strengthening pathways for community members while establishing an environmentally sustainable way to manage oyster shell waste (Figure 3). By addressing the environmental, health, and economic concerns, the project aimed to create a sustainable system that strengthens livelihoods.

Figure 3: Objective tree

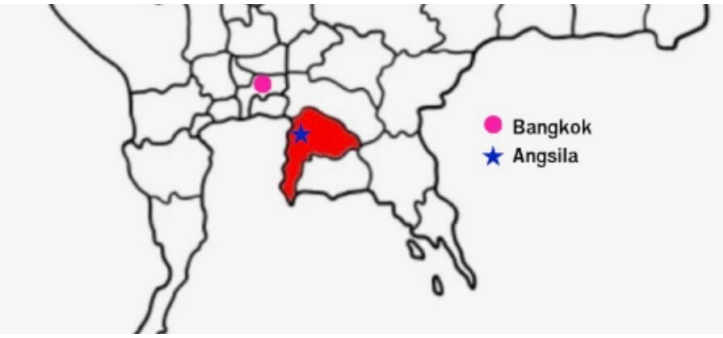


2 Literature Review

2.1 National Context

Thailand ranks as the world's 10th largest producer of mollusks, with mollusk farming serving as the third-largest contributor to Thailand's aquaculture industry. As of 2020, mollusks account for around 22% of the country's total aquaculture yield, with oysters among the dominant species in this group (FAO, 2024). Chonburi Province is Thailand's second-largest oyster-producing area, and Angsila is the province's largest oyster culture zone (Figure 4).

Figure 4: Chonburi is a province of Thailand, located in eastern Thailand



2.2 Background of Angsila

With a population of around 5,873, Angsila falls under the jurisdiction of the Angsila Town Municipality, which has a population of around 36,055 (CityPopulation.de, 2025). Angsila serves as a traditional fishing and seafood production center, maintaining its marine-based economy through small-scale fishing operations and coastal market activities throughout its history (Metalocus, 2023). The seafood market in Angsila operates as a popular destination for customers who want to purchase fresh oysters, mussels, and other shellfish from the surrounding coastal waters, which serve as the seafood hub of the area (Tourism Thailand, 2025).

The local economy depends on oyster harvesting operations and mussel shucking activities. The Department of Fisheries (2025) coastal shellfish culture survey identifies oyster farms as a vital component of Chonburi Province's aquaculture industry, which generates economic value. The operations of shucking houses and seafood vendors, and small-scale fishers, create financial opportunities for local households through their shellfish-related work.

Increased tourism creates ever greater demand for seafood products, as visitors seek to experience fresh seafood, coastal markets, and local dining. The demand for local consumption at restaurants and seafood markets, in turn, increases oyster and mussel shell waste (Wisdom Library, 2025). Thus, the combination of cultural heritage, the fisheries economy, and growing tourist demand makes Angsila an important location for oyster shell waste management research (FAO, 2003; Department of Fisheries, 2020).

2.3 Oyster Farming Practices in Thailand and Angsila

Oyster farming is a form of coastal aquaculture in which oysters are grown on human-made structures such as bamboo stakes, concrete poles, or hanging ropes placed in shallow marine environments (Figure 5, Figure 6). The goal is to provide oysters with suitable conditions such as stable salinity, clean water flow, and hard surfaces to support their attachment, growth, and eventual harvest (FAO, 2003).

Oyster farming is deeply woven into the local economy, culture, and identity. In total, a single oyster farmer averages a harvest of 100 bags of oysters every day. With an estimated 16 shucking houses in Angsila, this harvest produces around 35 tonnes of oyster shell waste every week. The growing demand for oysters has led to significant challenges with oyster shell disposal.

Thailand commonly cultivates several oyster species, including Crassostrea belcheri and Saccostrea cucullata. These species are some of the most commercially valuable and abundant along rocky shorelines in eastern provinces such as Chonburi. (Piyawongnarat et al., 2018; Department of Fisheries, 2020). Methods such as rope and rack culture remain popular because they require low investment and allow farmers to manage production year-round while adjusting to seasonal changes in growth rates and environmental conditions (Seachangjaew, 2023).

2.4 Oyster Shucking Activities in Angsila

Oyster production typically involves routine shucking at local markets and community shucking houses where oysters and mussels are prepared for sale to restaurants and tourists (FAO, 2003; Chooseangjaew, 2023). Although precise figures for the number of oysters shucked per day in Angsila are not publicly reported, provincial surveys and national shellfish statistics (Table 1) indicate that Chonburi is among the active shellfish production areas in the Gulf of Thailand, and that production shows clear seasonal variation related to breeding cycles and market demand (Department of Fisheries, 2020; Research on shellfish aquaculture in Thailand, 2000).

A small-scale, community-based local labor force that shucks and markets seafood is the basis for the oyster industry. Previous field studies in Eastern Thailand described strong female participation in small-scale oyster marketing and shucking activities and identified gendered labor patterns and household income from shellfish processing (Szuster, 2005). According to this study, female vendors obtained approximately 200 kg of oysters per week from local farms and paid an estimated 7 THB ($0.18 USD) per kg of oysters. Farm operations, as well as environmental monitoring of oyster farms in Angsila, are documented in local studies, and these reinforce shucking as a sustained community activity.

Figure 5: Angsila Oyster Scaffolding Pavilion

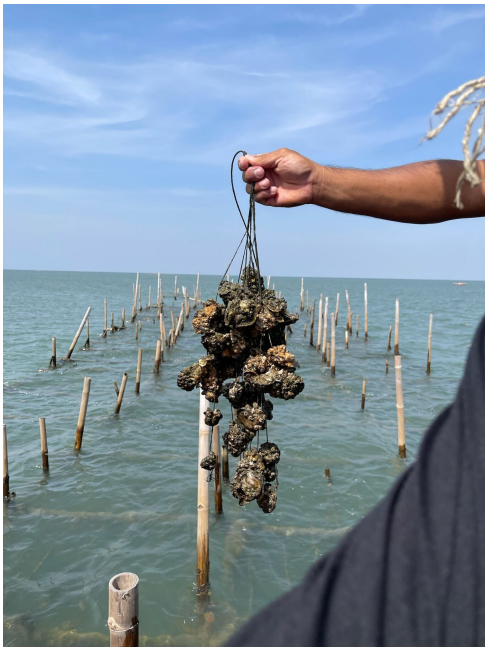


Figure 6: Bamboo-stake and hanging-rope oyster farming system

Table 1: Shellfish production by coastal zone and province in Thailand (tons).

Zone	Province	Total	Blood Cockle	Green Mussel	Oyster	Clams
Total	All Provinces	76,459.43	27,231.05	37,793.76	11,111.97	322.65
Zone 1	Trat	7,270.13	–	2,911.25	4,358.88	–
	Chanthaburi	221.65	–	–	221.65	–
	Rayong	1,163.01	–	903.59	259.42	–
Zone 2	**Chon Buri**	**5,177.01**	**769.86**	**2,096.92**	**2,310.23**	–
	Chachoengsao	803.45	–	803.45	–	–
	Samut Prakan	12,859.33	9,842.60	3,016.73	–	–
	Bangkok	754.98	457.13	297.85	–	–
	Samut Sakhon	4,940.47	4,419.81	510.66	–	10.00
	Samut Songkhram	20,664.63	4,556.91	15,885.91	–	221.81
	Phetchaburi	6,154.29	1,132.07	4,570.17	452.05	–
Zone 3	Prachuap Khiri Khan	194.81	–	119.70	75.11	–
	Chumphon	4,021.35	–	4,021.35	–	–
	Surat Thani	11,510.13	6,044.67	2,051.33	3,323.29	90.84
Zone 5	Ranong	567.74	–	567.74	–	–
	Phangnga	109.30	–	–	109.30	–
	Phuket	15.61	8.00	7.61	–	–
	Krabi	5.70	–	3.66	2.04	–
	Trang	25.54	–	25.54	–	–
	Satun	0.30	–	0.30	–	–

2.5 Problem Identification and Root Causes of Oyster Shell Waste Accumulation

The improper disposal of oyster shells creates various environmental problems that affect public health and social stability, particularly in coastal areas where shellfish processing is central to the economy. The shucking process leaves oyster shells with remaining organic materials, which include soft tissue and biofilm residues. The decomposition process of microbial organisms in open-air shell piles produces strong-smelling hydrogen sulfide (H_2S) gases through anaerobic reactions. The World Health Organization (2003) states that people who stay in areas with noticeable H_2S concentrations will experience respiratory problems and develop headaches and nausea. The strong odors from oyster shell waste decomposition create an urgent problem for coastal villages because they reduce the environmental quality and make the area less habitable for residents.

The decomposition of oyster shell waste produces two major water contamination threats. The passage of rainwater through shell piles leads to the release of dissolved organic compounds

and nutrients, which raise the biochemical oxygen demand (BOD) of nearby water bodies (U.S. Environmental Protection Agency, 2017). The presence of high BOD levels in water bodies reduces oxygen availability, which creates conditions for eutrophication that damages fish populations, benthic species, and aquatic plant life. The limited tidal exchange in Angsila's shallow coastal waters makes these environmental effects more severe, resulting in water stagnation, algae growth, and additional odor problems (Ma & Li, 2022). The continuous placement of shell waste near shorelines results in sediment changes and disrupts natural water flow patterns, which accelerates shoreline deterioration.

The management of shell waste without proper controls leads to multiple physical and visual environmental problems (Figure 7). The presence of large shell mounds creates environmental problems that negatively impact residents and make coastal tourist destinations less appealing.

Figure 7: Oyster Shell Waste Piles in Angsila, Thailand

The combination of shell waste with mud and debris in waterways blocks drainage systems, which increases flood risks when heavy rainfall occurs. The farming process leaves oyster shells with nylon ropes and plastic ties and farming materials, which break down into microplastics when exposed to sunlight and mechanical wear. The long-term ecological threats from microplastics exist because these materials stay persistent in the environment while building up in sediments and causing marine species to consume them, which worsens environmental damage (Ahmad et al., 2025). The environmental, social, and health problems in Angsila require an immediate solution through the implementation of a well-organized waste management system.

2.5.1 Challendges with Current Methods

Current frameworks provide limited incentives for compliant disposal, which contributes to the rampant improper dumping and continues to stimulate the improper shell waste cycle. Oyster-shell disposal in Angsila follows patterns commonly observed in small coastal seafood-processing communities across Thailand, where formal waste-handling systems for shell waste are limited. The piles are dispersed around numerous shucking houses, which makes it difficult to monitor and manage the waste. Observations from comparable Thai coastal towns show that shells mixed with residual soft tissue are frequently dumped directly along the shoreline, often forming open piles that accumulate over time. Some vendors and processors also use oyster and mussel shells as informal fill material for land reclamation or ground leveling, even though these shells often contain organic residues and attached farming materials such as nylon ropes, plastic ties, and fragments of bamboo stakes (Figure 8). These current practices reflect gaps in infrastructure and the absence of shell-specific collection mechanisms in local markets and shucking areas.

Figure 8: Oyster Shell Waste Piles with Atatched Nylon Rope in Angsila

2.5.2 Previous Initiatives

Previous efforts to manage and repurpose oyster shell waste have proven unsustainable due mostly to insufficient demand and the feasibility of production. For example, one project used

oyster shells to create bricks, but the process was not a cost-effective way of reducing sufficient amounts of shell waste. To address these limitations, our project looks to assess the market demand for our product to ensure that it is both feasible and sustainable. Some previous market chains for oyster shells, such as selling the shells to farms to use as feed and fertilizer, have been disrupted due to a change in oyster farming methods. When oysters were naturally farmed, they were able to be sold to oyster farmers. However, oysters are now harvested using concrete and nylon rope, which remains attached to the oyster shells, making them unsellable to farmers. With new cleaning methods that remove these additional materials, the old market chains can be reconstructed.

2.6 Solid Waste Policy and Regulations Pertinent to Oyster Shells

Thai law provides a clear framework for controlling oyster shell disposal and landfilling practices. Under the Public Health Act B.E. 2535, the collection, transportation, and disposal of solid waste are the responsibility of local authorities (พระราชบัญญัติการสาธารณสุข พ.ศ.2535). Oyster shells are considered solid waste; therefore, their disposal must be managed or approved by the municipality. Individuals or landowners are not legally allowed to create dumping or land-filling sites using oyster shells without authorization. In addition, any activity that involves accepting oyster shells in exchange for payment is considered a waste disposal service and requires a license from the local authority.

Local governments also have the power to prohibit waste dumping in public areas and to set hygiene standards for waste handling. This means that dumping oyster shells along roadsides, coastal areas, or vacant land, even for land-filling purposes, can be considered illegal. Furthermore, if oyster shells contain residual organic matter, their decomposition may cause odors, water pollution, or changes in soil quality, which may violate environmental quality regulations under the Enhancement and Conservation of National Environmental Quality Act B.E. 2535.

The Land Excavation and Land Filling Act B.E. 2543 further regulates the use of materials, including oyster shells, to raise land levels. Large-scale shell filling may require notification or approval from local authorities and must include proper drainage and safety measures. Improper land filling with oyster shells can lead to nuisance problems such as odor, dust, and water blockage. Together, these laws show that oyster shell disposal and land filling are regulated activities, yet improper practices regularly occur in coastal communities, which highlights the gaps between regulation and actual implementation (Appendix A).

2.7 Chemical and Mechanical Characteristics of Crushed Oyster Shells

To combat oyster shell waste, many researchers have investigated the physical properties of the shell when crushed. In the early 1980s, Japanese researchers executed their studies by taking three samples of the shells: non-crushed, crushed, and a mixture of crushed with other grains (Higashiyama et al., 2012). These samples were investigated for their permeability and compaction properties as a construction material, such as sand compaction piles.

Mechanical testing of soil mortar specimens incorporating varying proportions of crushed

oyster shell in place of sand revealed that up to 40% replacement did not significantly reduce compressive strength compared to normal cement mortar, highlighting the feasibility of partial substitution in construction mix designs (Yoon et al., 2003).

Similar to many other bivalves, the chemical makeup of oyster shells is comparable to limestone ($CaCO_3$); the typical chemical composition of oyster shells consists of 48.0-86.8% of CaO, which is the majority component of limestones (Mo et al., 2018). In addition, microstructural analyses show that the natural microstructure of the shells includes layered $CaCO_3$ crystals, contributing to their rigidity and chemical stability.

2.8 Oyster Shell Repurposing in Other Countries:

Japan: Oyster Shells for Cement and Construction Materials Japan has successfully integrated oyster shell waste into the construction industry by turning shells into calcium-rich additives for cement and concrete. Since oyster shells contain more than 90% $CaCO_3$, they can partially replace limestones in cement production, reducing raw-material extraction and lowering carbon emissions (Higashiyama et al., 2012). Japanese researchers have also developed fine powdered shell aggregates for use in environmentally friendly mortar and brick materials; the brick and mortar displayed high compressive strength and durability when compared with traditional materials (Yoon et al. 2003).

South Korea: Oyster Shells for Fertilizer and Soil Conditioner South Korea uses oyster shells as agricultural fertilizer and soil conditioners, especially in coastal farming regions. After cleaning and grinding, shells are used to raise soil pH, promote aeration, and supply calcium to crops like peppers and cabbage (Choi & Park, 2020). The Korean government encourages this technique by regulating oyster shells as recyclable resources rather than industrial waste, allowing farmers to get processed shell fertilizer at a low cost but heavily subsidized by the government.

United States: Oyster Shells for Coastal Restoration and Aquaculture In the United States, oyster shells are commonly utilized to restore natural oyster reefs and coastal ecosystems. The Chesapeake Bay Foundation and the Billion Oyster Project gather old shells from restaurants, cure them, and then use them as substrate for new oyster larvae (The Nature Conservancy, 2021). These regenerated reefs provide water clarity, biodiversity, and natural coastline protection at a high input cost.

2.9 Gap in Current Understanding

A perspective missing in many technical studies is the social and economic dimension of shell waste. Research from coastal communities in Vietnam, the Philippines, and Indonesia's coastal areas demonstrates that workers' waste management practices depend on their economic situation, unofficial work arrangements, restricted market areas, and presence or lack of local government oversight (Yoshioka et al., 2021). The cost of proper disposal and the absence of

financial incentives prevent informal seafood processing facilities from implementing proper shell separation, cleaning, and storage procedures. To protect their small profit margins, vendors and shuckers in Angsila focus on fast food preparation instead of waste management. Academic studies indicate that any shell recycling system needs to establish community partnerships, payment systems, and market-based rewards to achieve lasting behavioral changes (FAO, 2003; Chooseangjaew, 2023).

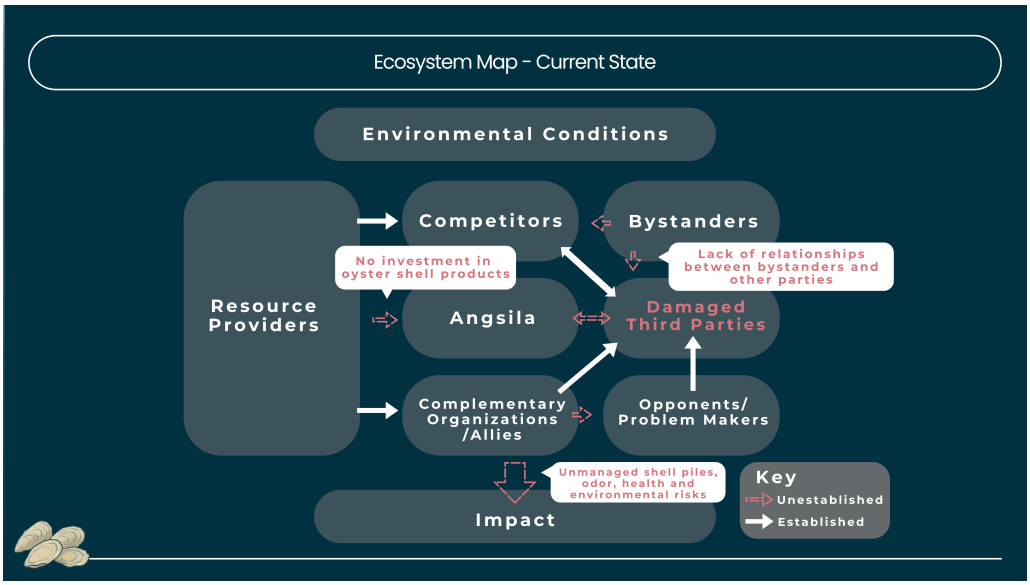
2.10 Ecosystem Map Current State

The figure below presents a simplified version of the current state ecosystem map (Appendix B). Ecosystem mapping is a social science tool that is utilized to show relationships and interactions between stakeholders in a system. The ecosystem map helps to identify gaps, conflicts, and opportunities for collaboration between stakeholders.

The current ecosystem map highlights the primary stakeholders in the oyster shell disposal ecosystem. These include oyster shucking houses, waste disposers, municipal authorities, and potential supporting groups. Currently, oyster shucking houses and waste disposers function as generators of shell waste, while the community and tourists are negatively impacted by the unmanaged shell piles, odor, and environmental impacts. The simplified map (Figure 9) shows how oyster shells flow through the ecosystem and how they impact stakeholders groups due to inefficient waste management.

The ecosystem map also demonstrates what missing from the current ecosystem including investment in oyster shell products, market interest, and collaboration. In the current ecosystem few few of these beneficial pathways exist which negativelt impacts the community. The ecosystem map serves as a valuable tool in indetifying where new relationships, and coordination mechanics are needed to support a beneficial circular ecosystem.

Figure 9: Ecosystem Map Current State Simplified



3 Methodology

The goal of this project was to propose a waste management system and a repurposing method for oyster shell waste in the Ang Sila Town Municipality. To achieve this goal, we developed the following objectives and strategies:

1. Identify gaps and potential capacities in the current oyster shell waste management system.
2. Develop a scalable model for how shells should be processed into raw materials for diverse value-added applications.
3. Analyze market feasibility and future scalability of the production model.

In this chapter, we will describe the methods and strategies that we developed to manage oyster shell waste and gather and analyze the data we collected from stakeholders. The conclusions from this data were used to develop our recommendations for waste-effective management.

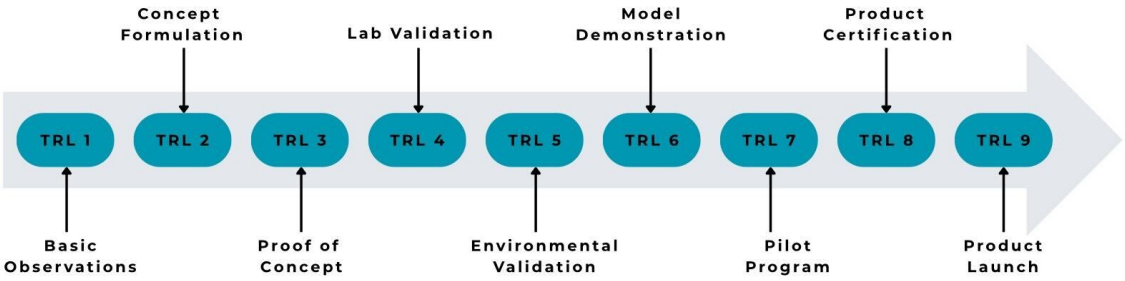
3.1 Objective 1: Identify gaps and potential capacities in the current oyster shell waste management system

To further understand the current conditions and impacts created by the waste piles, we reached out to many oyster shucking facilities and visited those who volunteered to participate in our study. Our visit to Angsila allowed us to witness the shucking process in action and gain insight into the direct experiences of the facility workers. Team members from Chulalongkorn University acted as translators during this process, with each interview lasting around 10 minutes. We were able to interview 24 oyster shuckers, and our questions from our interviews focused on gathering more detailed information on the disposal process and effects of the waste piles. We also asked questions regarding potential solutions and how our project could best cater to workers' needs. (See Appendix D)

3.2 Objective 2: Frame a structured model for how shells should be processed into raw materials for diverse value-added applications

We set out to develop a revised method that relocates the existing oyster shell piles away from residential areas and repurposes those oyster shells into raw materials that serve a benefit to the community. To structure the development of our repurposing process, we followed NASA's Technology Readiness Level scale (TRL) which is a nine-step formula for assessing the maturity of technology throughout its development (NASA, 2023) (Figure 10). Accordingly, the remainder of this section details the methods applied within TRL 1-4, encompassing basic observations, concept formulation, proof-of-concept development, and laboratory validation (See Appendix)

Figure 10: NASA Technology Readiness Level Scale (TRL)



3.2.1 Relocating Shells

To evaluate the feasibility of relocating the shells and establishing a centralized disposal location in small and large oyster shell shucking houses, we met with important officials in the community. Based on these meetings, we were able to compare our outside perspective of the problem with local expertise and community concerns (Figure 10). This comparison allowed us to synthesize a new solution to the oyster shell waste piles that exist in Angsila. Based on this analysis, we proposed two implementation scales for shell repurposing that align with community needs and operational feasibility:

Small-Scale Implementation: Oyster shell crushing would be managed directly by individual oyster shuckers. This approach enables shuckers to produce and sell crushed oyster shell (oyster shell powder) as a raw material to external repurposers. It minimizes infrastructure requirements while providing an additional revenue stream for small operators.

Industrial-Scale Implementation: Oyster shells would be collected at a centralized repurposing facility, where they would be crushed and processed into value-added products. This larger-scale approach allows for more efficient processing, product standardization, and the direct sale of finished goods to consumers. It also enables greater production capacity and long-term economic scalability.

These questions address the core concerns voiced in earlier interviews with the oyster shell shuckers and community officials. These questions factor into our proposed solution for managing the oyster shell waste.

3.2.2 Establishing a Central Disposal Site:

One of the core problems that exists in the effective management of the shells in Angsila is the lack of a central disposal site. Therefore, establishing a central disposal site that can accommodate the estimated thirty-five tonnes a week that is generated by the municipality is a priority. This site must be selected with the input of the municipality in order to minimize the negative impacts that current practices create. As a part of this practice, all shucking houses

must be encouraged to provide their oyster shells to community collection practices and not private collectors that exist today (Figure 11).

Based on interviews with Angsila locals, many residents living near the piles have expressed an interest in improving shell waste management but reported limited knowledge regarding technical expertise and how to approach the piles. On the other hand, others do not consider shells as a potential source of revenue, but rather as waste. These perspectives indicate the importance of raising awareness of the value of oyster shells to gain their engagement in shared disposal sites.

Figure 11: Oyster Shell Waste Pile at a Large Shucking House

3.2.3 Seperating the Nylon Rope

Current farming practices generate oyster shells that are intermixed with nylon rope and concrete (Figs. 13, 14). These byproducts prohibit the shells from being easily used for certain

applications.

Figure 12: Oysters on Nylon Rope

Figure 13: Nylon Rope from Shucking House

To effectively separate the nylon rope from our oyster shells, the oyster shells must be crushed. Oyster shells can be crushed on a small scale using a jaw crusher or on an industrial scale using a coarse shell crushing system, depending on the desired processing capacity. The different crushing machines allow for scalable implementation of the repurposing process.

Figure 14: Jaw crushing machine

3.2.4 Cleaning the Crushed Shells

After the nylon rope is separated from the oyster shells, the concrete is the next farming byproduct that must be separated from the oyster shells. To separate the concrete from the oyster shells, as well as further sanitize the oyster shells, the shells were placed in a water bath and underwent agitation for sixty minutes to separate residual meat from the shells. Next, the shell fragments and concrete were sieved to separate the concrete from the shells. This process resulted in oyster shell fragments and concrete being separated into two distinct layers.

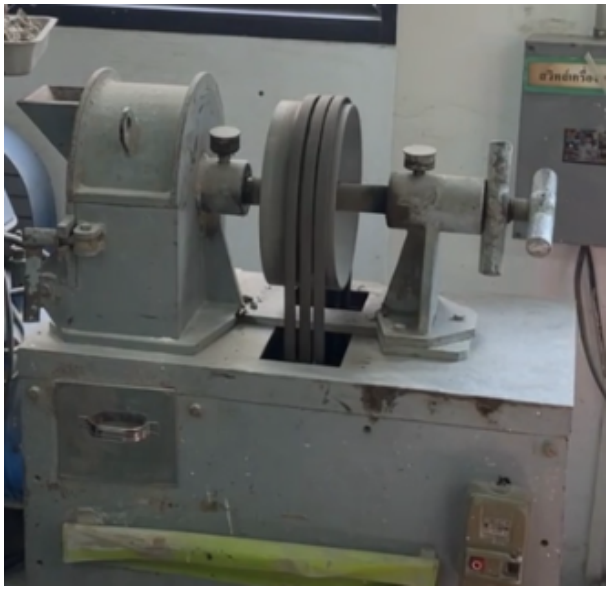
3.2.5 Drying the Shells

The shells were then crushed and underwent the cleaning described above, the shells underwent drying to further sanitize them. This drying process lasts around 3-5 days and involves spreading the shells over well-ventilated areas or outdoors with minimal piles in order to properly dry the crushed shells and prepare the shells for grinding.

3.2.6 Grinding the Shells

Shells then underwent a grinding process to make the shells into usable particulates that are easily storable in containers such as barrels (Figure 15). Similar to the industrial crushing machine (Figure 14), the fine grinder machine allows for grinding of coarse oyster shells into a fine powder that can be utilized in a variety of value-added products

Figure 15: Fine grinder machine



3.2.7 Prototype Production

Pavers Prototype Production Method To make a paver, instead of using oyster shell powder, coarse ground shell (less than 5 mm in size) was mixed with cement.The mixture was prepared with the following ratio: 1 kilogram of cement, 3 kilograms of coarse ground shell, and 0.6 kilograms of water. This portion was sufficient to produce 2 pavers. The mixture was then placed into paver molds (Figure 16), and a hydraulic press was used to compress the mixture into shape, ensuring the pavers had the required strength. By incorporating coarse ground shells in the mix, more oyster shell waste was utilized, significantly contributing to waste reduction and promoting a more sustainable approach to material usage

Figure 16: Compression Mold for Paver Formation

Tile/Coaster Protoype Production Method For the coaster prototype, oyster shell powder was combined with cement and water at a ratio of 4:2:1, respectively, to produce a workable mixture suitable for molding. The mixture was poured into molds to achieve the desired shape. A hydraulic press (Figure 17) applying 50 MPa of pressure was then used to compress the

material, enhancing its density and mechanical strength. The molded product was air-cured for 1 day, followed by water curing for 3-4 days to further improve strength development. This process is consistent with the method used for tile production. Therefore, it can be readily adapted for future tile manufacturing by modifying the mold design during the shaping stage.

Figure 17: Hydraulic Press for Coaster and Tile Production

3.2.8 Lab Testing

Two possible products based on ground oyster shells, coasters, and pavers, were developed and tested. Their material properties were evaluated through two tests: absorption and flexural strength.

Water Absorption Test The water absorption test involves several steps to determine the amount of water absorbed by a material.

1. Immerse the sample in water for 1 hour.
2. Remove the sample from the water and gently wipe the surface with a damp cloth to remove any excess surface water.
3. Weigh the sample to obtain its wet weight.

4. Dry the sample in an oven at 60°C for 24 hours (1 day). Once dried, weigh the sample again to obtain its dry weight.

5. Calculate the water absorption percentage using the following formula:

$$\text{Water Absorption (\%)} = \frac{\text{Wet Weight} - \text{Dry Weight}}{\text{Dry Weight}} \times 100$$

3.2.9 Establishing a Central Disposal Site:

One of the core problems that exists in the effective management of the shells in Ang Sila is the lack of a central disposal site. Therefore, establishing a central disposal site that can accommodate the estimated 7 tons a week that is generated by the municipality is a priority. This site must be selected with the input of the municipality in order to minimize the negative impacts that current practices create. As a part of this practice, all shucking houses must be encouraged to provide their oyster shells to community collection practices and not private collectors that exist today(Figure 11).

subsection Removing Odor The foul odors emanating from the oyster shells remain one of the community's primary concerns. A possible solution to reduce these odors was developed, utilizing photosynthetic bacteria, microorganisms, RO water, and calcium hydroxide to remove residual organic matter from oysters (Table **??**). This method required testing by participants in order to identify the best mix of photosynthetic bacteria inoculum, EM inoculum microorganisms, water, and calcium hydroxide. The problem with testing for odor is that it is primarily a subjective-based measurement relying on human senses. Therefore, participants were asked to rank the different solutions added to oyster shells to identify the most preferred mixture of bacteria and microorganisms. A blind test was performed to eliminate treatment allocation bias for the various tanks. The buckets were placed in numerical order and were not varied between participants.

The experiment for odor treatment of oyster shells with microorganisms involved cleaning the fresh oyster shells with plain water and drying them under the sun for one day.

1. A 1-kilogram sample of oyster shells was weighed and placed in a 2-liter plastic container. Water was sprayed evenly over the shells with a total of 200 grams of water.

2. The shells were then sprayed with a solution of calcium hydroxide and/or microbial inoculum, as described in the table:

    - Tank 1: sprayed only with water.
    - Tanks 2, 5, 6, and 7: sprayed with a solution of calcium hydroxide.
    - Tank 5, 6, and 7: after spraying with calcium hydroxide solution for about 1 hour to adjust the pH to neutral or slightly alkaline, the solution was drained, and then the microbial inoculum was sprayed evenly over the shells.

- Tank 3, 4, and 8: sprayed directly with the microbial inoculum solution without adjusting the pH.

3. The containers were sealed and placed in the laboratory for storage. Odor testing was conducted on the 7th day, and the pH of the solution in each container was also tested.

Tank #	H_2O (ml)	Calcium Hydroxide (ml)	Photosynthetic Bacteria (ml)	Effective Microorganisms (ml)
Tank 1	200	0	0	0
Tank 2	200	10	0	0
Tank 3	200	0	10	0
Tank 4	200	0	0	10
Tank 5	200	10	10	10
Tank 6	200	10	10	0
Tank 7	200	10	0	10
Tank 8	200	0	10	10

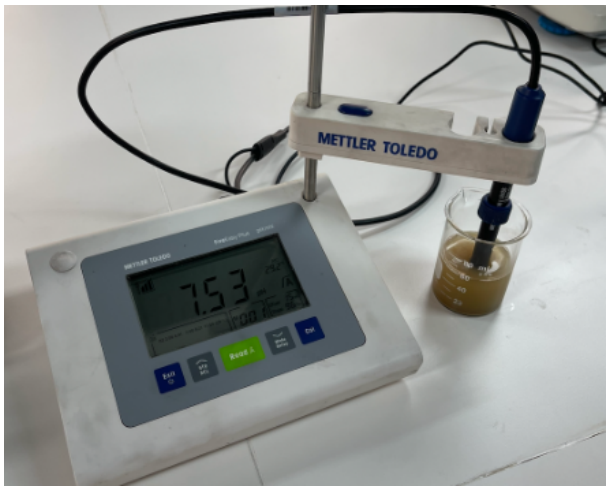
Table 2: Composition of treatment solutions used in oyster shell odor mitigation experiments.

Participants were then asked to rank these solutions from worst [a score of one] to best [a score of eight]. These participants represented a wide range of backgrounds, allowing for cultural preferences to be considered. Background factors must be considered in this research, as the new odor must appeal to residents and tourists. Tourism was identified as a crucial industry in the interviews referenced above; the smell preferences of the residents who must deal with the odor daily, and tourists must be taken into account.

pH Testing Process The pH of the liquid collected from each treatment tank was measured following the completion of the odor testing experiment. The water in each tank (Figure 18), which had been in contact with the oyster shells during the odor treatment process, was sampled to assess changes in acidity or alkalinity resulting from the applied treatments. A calibrated digital pH meter (Figure 19) was used for all measurements. Before testing, the instrument was calibrated using standard buffer solutions to ensure accuracy. The pH electrode was immersed directly into the collected water sample. The pH value was recorded once the reading stabilized. All measurements were conducted at room temperature, and each tank was tested under consistent conditions to allow for comparison among treatments.

Figure 18: Water Sample Collected from Treatment Tank for pH Measurement

Figure 19: pH Measurement Using a Digital pH Meter



Flexural Strength Test To conduct the flexural tests, five samples were produced for each oyster shell percentage mixture (0%, 25%, 50%, and 75%), with 0% serving as the baseline. Each sample had dimensions of 30 mm in width, 60 mm in length, and 5 mm in thickness to allow proper testing (Figure 20).

The samples were placed on a three-point bending testing platform where an increasing load was applied at the center of the specimen until fracture occurred (Figure 21). The breaking load recorded during the test was then used to calculate the flexural strength using Equation 1.

$$M = \frac{3PL}{2bd^2} \tag{1}$$

In this equation, P represents the breaking load, L is the distance between the supports, b is the average specimen breadth, and d is the average specimen depth. The equation calculates the modulus of rupture (flexural strength) for each sample. After calculating the flexural strength for all samples, the values for each oyster shell percentage were averaged.

Figure 20: Rectangular Specimens Prepared for Three-Point Bending Test

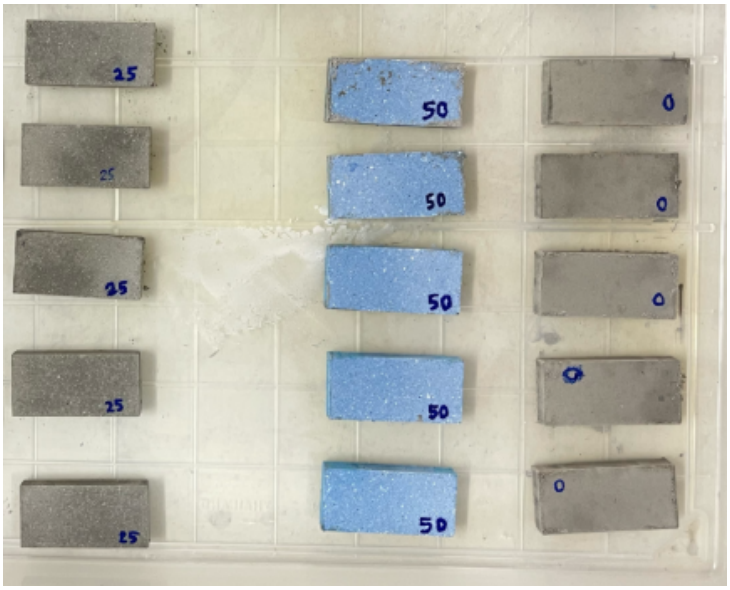
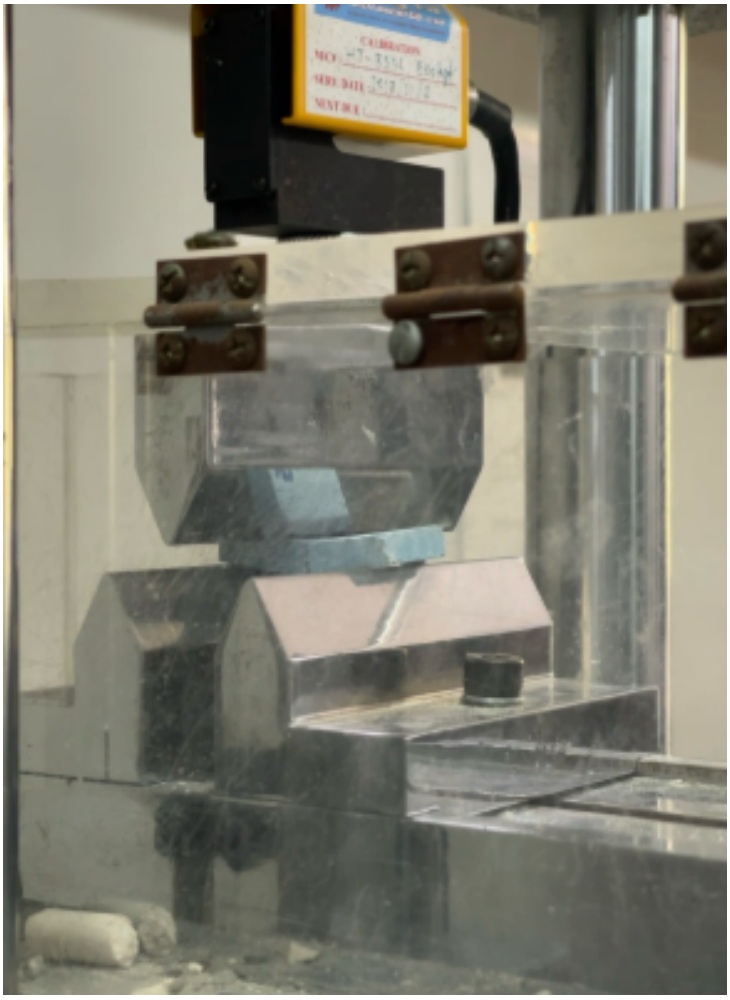


Figure 21: Three-Point Bending Test



3.3 Objectve 3: Analyzing Market Feasability and Future Scalability

3.3.1 Market Analysis

We conducted a market analysis to assess market demand, identify target customers, and explore potential opportunities. We collected market data through primary research, which involved interviews and surveys with oyster farmers, and associated businesses, as well as secondary research sourced from government documents, scholarly research, and industry reports. We analyzed the volume of oyster shell waste generated, existing waste management practices, and stakeholder interest in alternative uses of oyster shells (Figure 22). Qualitative analysis and fundamental market segmentation were used to evaluate market feasibility and opportunities for sustainable solutions.

Figure 22: Oyster Shell

3.3.2 Analysis of Inputs

To analyze the cost of the oyster shell cleaning and repurposing process at both small and large scales, we used the same process. We evaluated two operating scenarios: a small-scale pilot model that relies on manual or low-tech equipment (basic crushing, washing tanks, manual sieving, air drying, and small grinders) and a large-scale mechanized model that uses machinery for higher throughput (mechanical crushers, mechanical separators, washing systems, controlled drying, and industrial grinding). All costs were calculated using one tonnes of incoming shell

waste as a consistent base unit, allowing direct comparison across scales. We computed the total processing cost per tonnes and converted it into cost per kilogram of finished shell powder, enabling us to compare cost efficiency between manual and mechanized options.

4 Results and Analysis

In order to address the problem of oyster shell waste management, we initially conducted interview with key stakeholders including municipal officers, oyster shuckers, and oyster farmers. In response to the concerns of stakeholders, we devised a processing method to effcientely collect, clean, and crush oyster shells for repurposing. Finally, we devloiped a market analysis to evaluate potential applications for processed oyster shell and evaluate there economic feasibility.

4.1 Interviews Identified Core Requirements for Managing Oyster Shell Waste

4.1.1 Key Findings from interviews with oyster shucking houses

Strong odors and fruit fly infestations stem from leftover oyster meat on the shells. These problems are exacerbated by the large quantity of waste generated, leading to challenges in managing the shells effectively and maintaining cleanliness and hygiene in the workplace. When asked, the village head indicated that if there is a clear approach and strong potential for success, they would be willing to organize and bring the community members together to participate in the activities we devised (Appendix D). This shows openness and a desire for collaboration, provided the initiative demonstrates real benefits for the community.

4.1.2 Angsila Municipality

The proper disposal of oyster shells rests with the owners of the oyster shucking facilities. Oyster farms only deliver oysters to the shucking areas, and the shells remain at the facility, as restaurants typically purchase only the meat, not the shells. The growing waste problem presents ongoing challenges because the municipality's existing landfills for the disposal of these shells are largely filled, so oyster shells are illegally dumped in other locations. The village head indicated that if there is a clear approach and strong potential for success, they would be willing to organize and bring the community members together to participate in the activities we devised. This shows openness and a desire for collaboration, provided the initiative demonstrates real benefits for the community.

4.2 Environmental & Health Impacts

Workers expressed minimal concern regarding the health and environmental impacts of the shells, identifying foul odors as their primary issue. These odors result from residual organic matter that decomposes on shells that are left in the sunlight (Figure 23)espite the community's concerns, the oyster piles continue to grow in size due to the demand for oyster meat from residents and tourists. This perpetuates the problem for the municipal city as it must manage consumer demand with local capabilities. Thus, the municipality has begun to explore improvements in current disposal methods to address evolving farming practices and the growing

demand for oyster meat.

Current farming practices for oyster shells utilize nylon rope to bundle shells and support larvae attachment, and concrete to improve the oyster yields (Figure 23,Figure 24). While these practices benefit the oyster market system, they contribute to a negative environmental impact. However, Angsila Municipality is presently working on combating plastic waste, so utilizing the current systems creates an effective foundation for managing the nylon.

Figure 23: Large Oyster Shell Piles in Angsila, Thailand

Figure 24: Fresh oysters collected from bamboo stakes

While the current municipal systems cannot be directly applied to oyster shell waste management, they provide guidance for increasing community involvement and participation in responsible waste management practices. To manage the existing waste collection, organizations involved in plastic waste collection should be utilized to relocate waste away from residences.

While improved collection and relocations are crucial for the waste piles, foul odor remains a critical part of creating a comprehensive solut

4.3 Collection and Management

In addition to odor concerns, interviews identified that another concern amongst the community is the management of oyster shell waste. The current disposal methods encourage improper disposal and ineffective collection of shells. To maintain odor control, the establishment of a central collection site is essential. This would reduce the amount of investment required to deploy the photosynthetic solution and contribute to coordinated waste handling. The collection site should be selected by the municipality to reduce impact on residents and future developments (Figure 25). This ensures that community concerns are considered when selecting the proper location for managing future oyster shell waste.

Figure 25: Angsila Village Head visits the oyster shell piles

Addressing the growing oyster shell crisis requires an investment of resources and time. This investment from the municipality will take the form of a financial investment in establishing a central collection location and collection of existing oyster shell piles on beaches and in marine environments. While the financial investment by the municipality may seem like an undue burden to place upon the local community, the initial cost of this site will be offset by:

- Reduced odor and improved public health conditions
- Cleaner beaches and marine environments
- Development of a value-added oyster shell repurposing industry

The central location should integrate existing oyster shell waste collectors to ensure compliance with proper disposal practices and minimize negative externalities. Collaboration between current collectors in centralized locations improves the removal of shell waste from residential

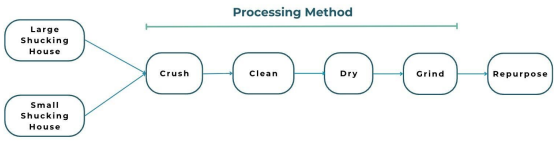
areas. The location could also serve as a hub for repurposing operations where oyster shell waste can be converted into value-added products using the previously described methods (Figure 26).

Figure 26: Angsila Village Head visits the oyster shell piles

4.4 Processing Method

Analysis of stakeholder input, site conditions, and processing feasibility suggests that oyster shell repurposing is viable at both the small scale and the industrial level. The processing framework in (Figure 27) outlines the proposed oyster shell waste repurposing model that can be deployed in scalable phases in the municipality.

Figure 27: Angsila Village Head visits the oyster shell piles

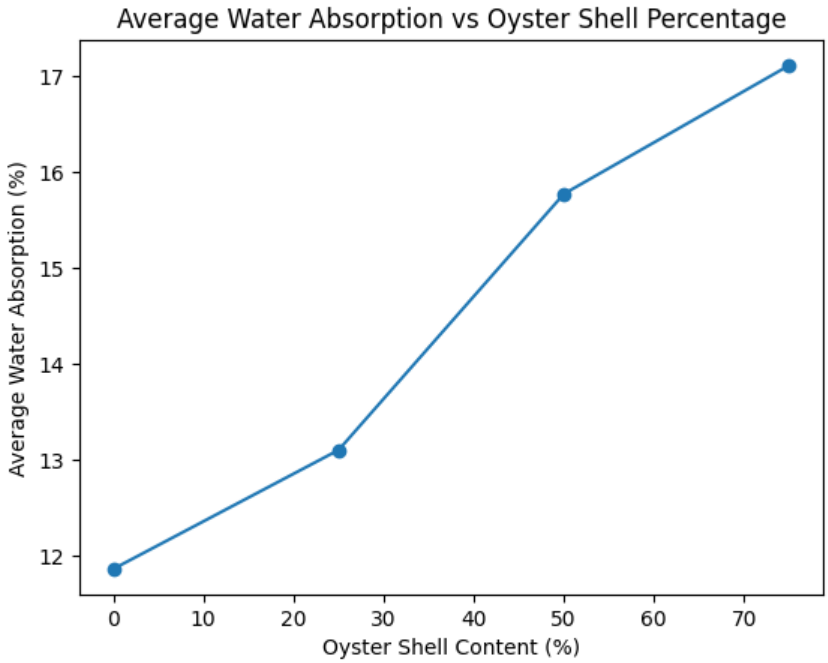


4.5 Material Properties

The water absorption test is a critical procedure in evaluating the performance of construction materials, particularly for products like pavers and tiles. This test helps determine the material's ability to absorb water, which directly impacts its durability, strength, and overall quality. Materials that absorb too much water may weaken over time, leading to issues such as cracking, erosion, or reduced structural integrity (Figure 28). By conducting the water absorption test,

manufacturers can assess how well their materials will perform in various environmental conditions and ensure that they meet the required standards for construction use. It also provides valuable insights for improving the material's formulation, helping to enhance its long-term reliability and suitability for different applications.

Figure 28: Water Absorption vs Shell Percentage



The results above show the applicability of oyster shell + concrete for absorbing water, which allows for certain applications of oyster shell products. While absorption is an important measure for how effective oyster products are at collecting water that can then later be evaporated over time, other properties were also evaluated to determine the effectiveness of these materials. Three-point bending tests were conducted on different mixtures of 0%, 25%, 50%, and 75% oyster shell to determine the flexural strength of oyster shell + concrete mixtures. 0% concrete was the control factor in these tests, and a linear relationship between the amount of oyster shell and decreasing flexural strength was observed (Figure 29,Figure 30).

Figure 29: Oyster Coaster and Tiles of Different Ratios

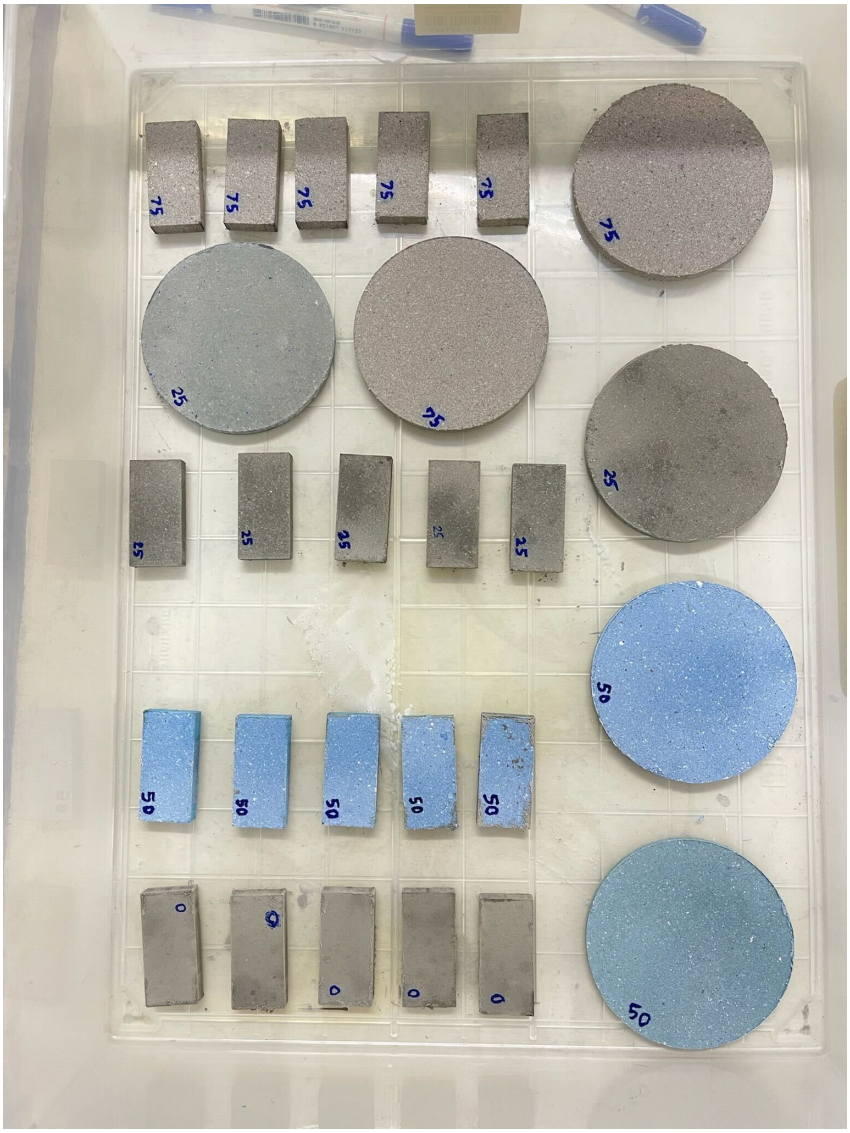
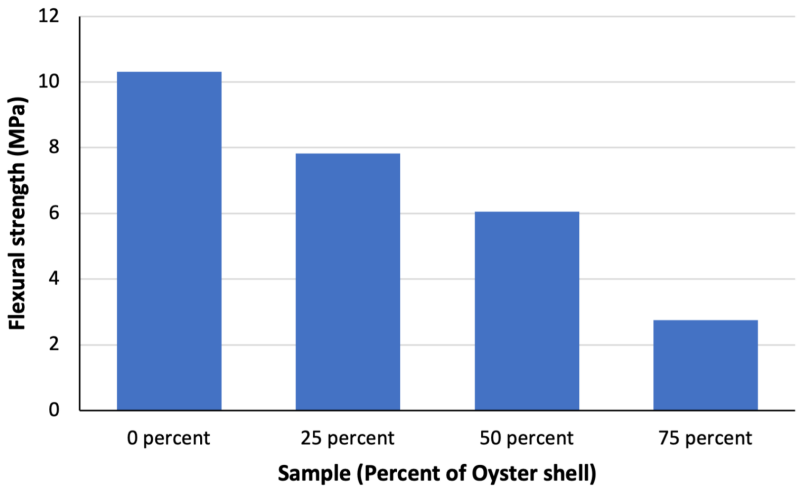


Figure 30: Flexural Strength vs Oyster Shell Percentage Bar Graph



The results show that the 0% oyster shell sample exhibited the highest flexural strength at approximately 10.31 MPa, followed by 25% at about 7.82 MPa and 50% at around 6.01 MPa,

while the 75% sample showed the lowest strength at approximately 2.75 MPa. Based on this trend, it can be concluded that as the percentage of oyster shells increases, the flexural strength of the material decreases progressively.

The amount of oyster shell determines the usability of the oyster shell products for different industries. Based on the relationship between the amount of oyster shell and absorption. and flexural strength, the feasibility of using oyster shells in different concrete products can be evaluated.

4.6 Repurposing

When coming up with repurposing methods, it is crucial to take into consideration the demand, safety, feasibility, and cost associated with each product. We determined that the best way to repurpose the shells is by selling them as the raw, ground-up material. Since raw oyster shell powder showed the broadest demand, it can be sold to multiple and diverse industries such as agriculture and construction. In addition, value-added products such as coasters, pavers, and tiles have been selected based on their material requirements and alignment with the composition of oyster shells.

Oyster shells offer unique advantages due to their material properties discussed above, as well as their chemical makeup. These advantages allow for oyster shell products to be competitive and offer greener solutions to current practices across industries. Coasters were identified for use with oyster shell products due to oyster shells' ability to alter the porosity and absorbency of concrete coasters. Tiles were selected because their production allows for high volumes of shells to be incorporated, making them effective for reducing shell waste. In addition, the main component in oyster shells is calcium carbonate, which is a material commonly used for construction applications.

Pavers are also a potential repurposing application due to their ability to utilize large numbers of shells. Pavers require a high porosity and permeability to reduce flooding risk, particularly in urban environments. Due to this, oyster shell pavers offer some unique advantages compared to current concrete pavers. Oyster shell pavers, as evidenced above, can absorb larger quantities of water when compared with traditional concrete pavers.

Another set of repurposing applications is the applicability of oyster shell powder in agricultural products. Due to the oyster shells' chemical makeup of calcium carbonate, oyster shell powder can be utilized in fertilizer and animal feed. In fertilizer, oyster shell powder can serve the same purpose as agricultural lime. The weakness of oyster shell powder compared to agricultural lime is that it possesses little to no magnesium, which poses a problem for its utilization as a replacement for agricultural lime. A more practical application for oyster shell powder in fertilizer is as a calcium supplement in fertilizer blends. Oyster shell powder would be utilized to increase the calcium content of soil, though it does not possess sulfur for crops requiring increased sulfur content. Another agricultural application for oyster shell powder is as a calcium supplement for animal feed, particularly in poultry. Chumkam and Jintasaporn found that calcium supplementation improved eggshell quality and productive performance in laying hens (Chumkam et. al., 2020). Therefore, oyster shell powder serves as a way to increase profit for

farmers while serving as a waste management system for aquaculture.

4.7 Consumer Markets

As mentioned previously, the desires of the stakeholders are to develop a marketable end product that contributes to the economy of the municipality. The previous farming method that did not utilize concrete in the harvesting of oyster shells allowed for whole shells to be sold to farmers as calcium supplements that strengthened the quality of eggs. The current method that utilizes concrete to firmly attach the shell to ropes makes the oyster shells unusable for farms. We propose utilizing the nascent market for oyster shells again by selling the crushed and ground oyster shell particulate as a new calcium supplement. This new supplement will revitalize the market that has been abandoned by advances in oyster farming methods while contributing to the circular economy of Angsila. Oyster shell powder also has applications in the construction industry as an aggregate. We propose utilizing the material properties of calcium carbonate in the oyster shells to construct tiles and bricks for decorative and non-structural construction.

Furthermore, the crushed oyster shell powder has a variety of other uses that can be tapped into as secondary revenue streams for the project. Oyster shell powder can be utilized in cosmetic products such as body scrubs. With further cleaning, such as a bleach bath, the oyster shell powder can be sold as an abrasive in beauty care products. The crushed oyster shell powder can also be utilized in fertilizers with no additional treatment as a way to increase the lime content of fertilizer blends, decreasing the pH level of the fertilizer.

The additional revenue sources for the oyster shell powder allow for the producers of the powder to explore additional revenue streams while relying on a stable revenue from a revitalized local economy for calcium supplements. We envisage these additional revenue sources as being critical for the long-term viability of the oyster shell waste repurposing project.

4.8 Cost Analysis

The cost analysis in this study is conducted based on key cost components relevant to industrial-scale production. The calculation framework includes raw material costs, production and processing expenses, labor costs, machine and equipment investment, operational energy consumption, and other related overhead costs. At this stage, detailed cost data are available only for the coaster (business-to-consumer(B2C)) and brick (business-to-business(B2B)) products.

The cost analysis for oyster shell-based coasters was developed under defined production assumptions reflecting industrial-scale operations. Each coaster has a diameter of 10 cm and an average weight of 100 grams per piece. The material composition consists of approximately 70% crushed oyster shell powder (particle size <1 mm) and 30% white cement as a binder (Figure 31). The projected production capacity is 200,000 pieces per year, approximately 1,000 pieces per day, requiring around 20,000 kilograms of oyster shells annually. Raw oyster shells are assumed to be procured as waste materials with a management and handling cost of approximately 30,000 THB per year (Table 3).

Figure 31: Dimensions, Weight, and Makeup of Our Oyster Shell Coaster

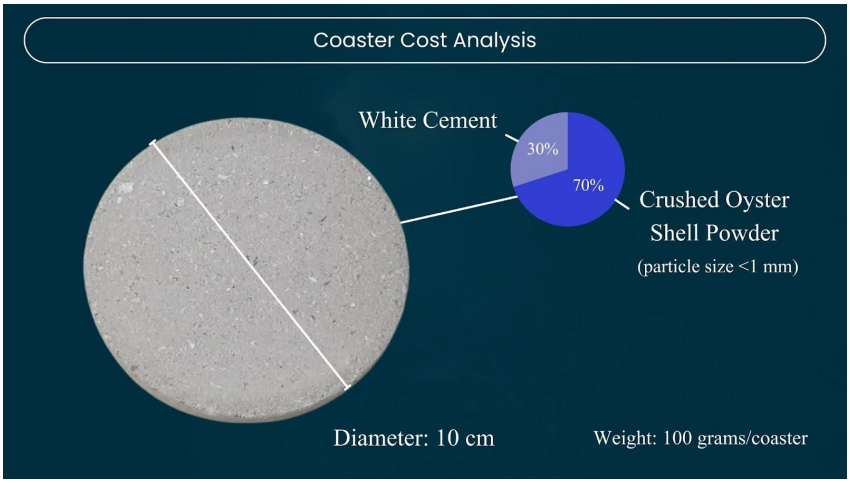
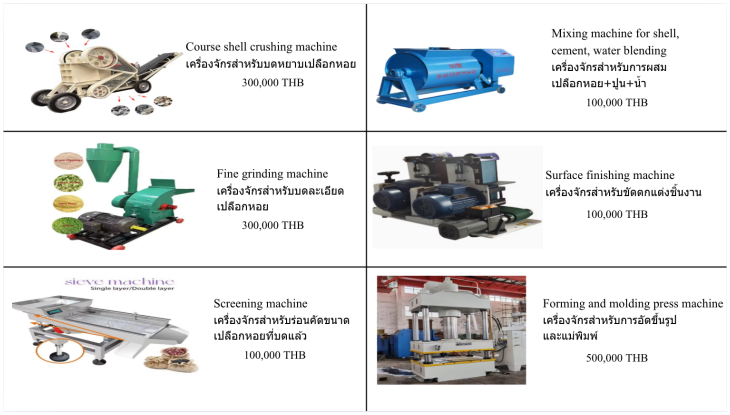


Table 3: Cost Breakdown for Oyster Shell Coasters

Category	Item	Annual Cost (THB)
Material Preparation & Processing	Shell collection & transportation	10,000
	Labor for washing, microbial cleaning & sorting	10,000
	Cleaning materials	10,000
Machinery Depreciation (7-year)	Coarse shell crusher	42,857
	Fine grinder	28,571
	Screening machine	14,286
	Mixing machine	14,286
	Hydraulic brick forming & molding machine	71,429
	Surface finishing machine	14,286
Utilities (Production)	Electricity for machinery operation	70,000
Direct Labor	4 workers (mixing, molding, finishing, packaging)	320,000
Packaging Costs	Paper boxes	400,000
	Plastic bags	50,000
	Stickers	200,000
Administrative & Overhead	Area rental	400,000
	Maintenance	200,000
	Quality control	50,000
	Administrative costs	400,000
Totals	**Total annual production cost**	**2,495,914**
Cost Estimated Per Coaster		**15**

Based on this cost structure, proposed selling prices vary by market segment to ensure sufficient profit margins. For wholesale markets, the recommended selling price ranges from 25-30 THB per piece, generating estimated annual revenues of 2-3 million THB. For retail markets, prices range from 35-50 THB per piece, with projected annual revenues of 4-7 million THB. For tourist souvenir markets, premium pricing between 45-59 THB per piece could generate annual revenues of approximately 6-8.8 million THB. Overall, the analysis indicates that oyster shell coasters are economically feasible under industrial-scale production, with strong margin potential across different distribution channels (Figure 32).

Figure 32: Potential Repurposing Machinery & Cost for Coasters



The cost analysis for oyster shell-based pavers was also developed under industrial-scale production assumptions to evaluate economic feasibility in the construction B2B market. Each paver has dimensions of 10 cm × 20 cm × 6 cm, with an average weight of approximately 2.0 kilograms per paver. The material composition consists of 75% crushed oyster shell and 25% Portland cement as a binder (Figure 33). The projected production capacity is 500,000 pavers per year (approximately 2,500 pavers per day), requiring around 750,000 kilograms of oyster shell waste annually. The estimated shell management and handling cost is approximately 250,000 THB per year (Table **??**).

Figure 33: Dimensions, Weight, and Makeup of our Oyster Paver

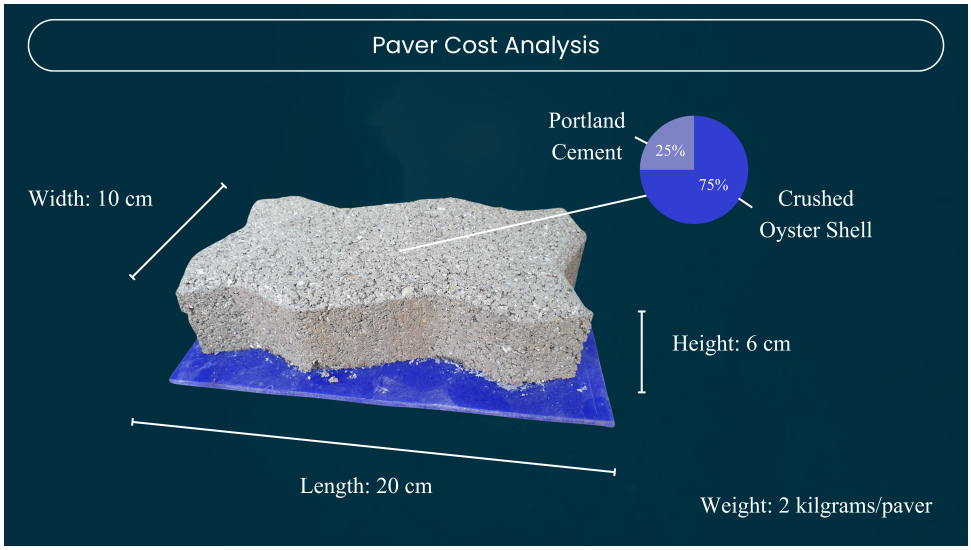
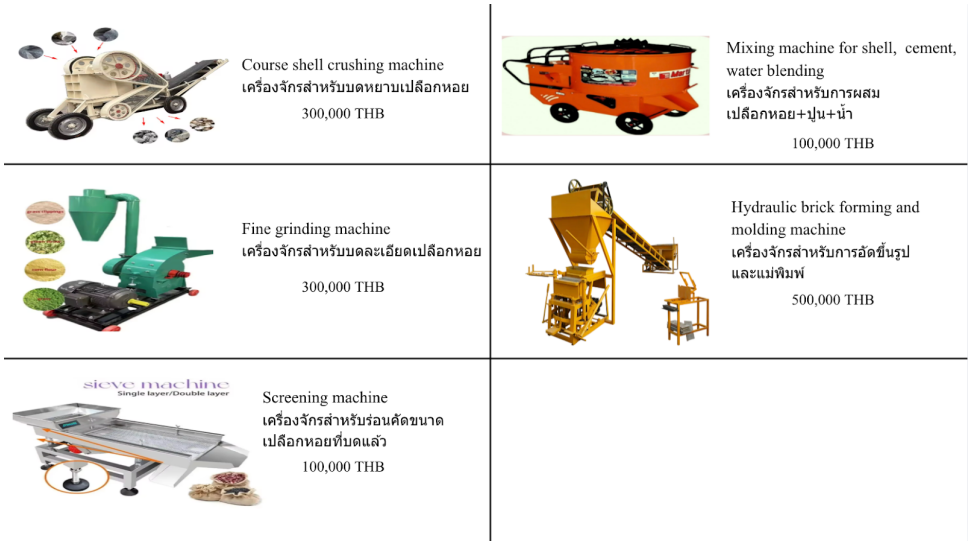


Table 4: Cost Breakdown for Oyster Shell Coasters

Category	Item	Annual Cost (THB)
Material Preparation & Processing	Shell collection & transportation	10,000
	Labor for washing, microbial cleaning & sorting	10,000
	Cleaning materials	10,000
Machinery Depreciation (7-year)	Coarse shell crusher	42,857
	Fine grinder	28,571
	Screening machine	14,286
	Mixing machine	14,286
	Hydraulic brick forming & molding machine	71,429
	Surface finishing machine	14,286
Utilities (Production)	Electricity for machinery operation	70,000
Direct Labor	4 workers (mixing, molding, finishing, packaging)	320,000
Packaging Costs	Paper boxes	400,000
	Plastic bags	50,000
	Stickers	200,000
Administrative & Overhead	Area rental	400,000
	Maintenance	200,000
	Quality control	50,000
	Administrative costs	400,000
Totals	**Total annual production cost**	**2,495,914**
Cost Estimated Per Coaster		**15**

Based on this cost structure, the recommended wholesale selling price ranges from 8-10 THB per unit, generating an estimated annual profit of approximately 1-2 million THB, or a margin of approximately 2-4 THB per paver. Overall, the analysis indicates that oyster shell-based pavers are economically competitive with conventional pavers, while offering additional environmental value through waste utilization and circular economy integration, making them attractive for large-scale construction and public infrastructure projects (Figure 34).

Figure 34: Dimensions, Weight, and Makeup of our Oyster Paver



4.9 Market Analysis

The market analysis is divided into two segments: business-to-business (B2B) and business-to-consumer (B2C), as oyster shell-based products address different customer needs and value propositions. The B2B market focuses on functional, volume-driven applications such as construction materials and agricultural supplements, while the B2C market targets consumer products where sustainability, aesthetics, safety, and brand storytelling play a key role in purchasing decisions. Separating these markets allows for clearer evaluation of feasibility, regulations, and value creation. Accordingly, this study applies marketing strategies, including 4Ps analysis, SWOT analysis, and product positioning, to systematically assess market potential and define appropriate strategies for each segment.

4.9.1 B2B: Tiles and Pavers

Our analysis highlights a value proposition focused on cost efficiency and sustainability. The focal products are pavers and tiles, where oyster shell powder serves as a calcium carbonate ($CaCO_3$) substitute while meeting construction material standards.

- The pricing strategy focuses on cost competitiveness through locally sourced raw materials, reduced transportation expenses, bulk production, and long-term supply contracts with manufacturers.
- Distribution is conducted through direct B2B channels targeting tile and brick manufacturers, construction material suppliers, real estate developers, and municipal infrastructure projects.
- Promotion focuses on technical performance, compliance with standards, pilot project demonstrations, and alignment with green building and ESG policies rather than consumer branding.

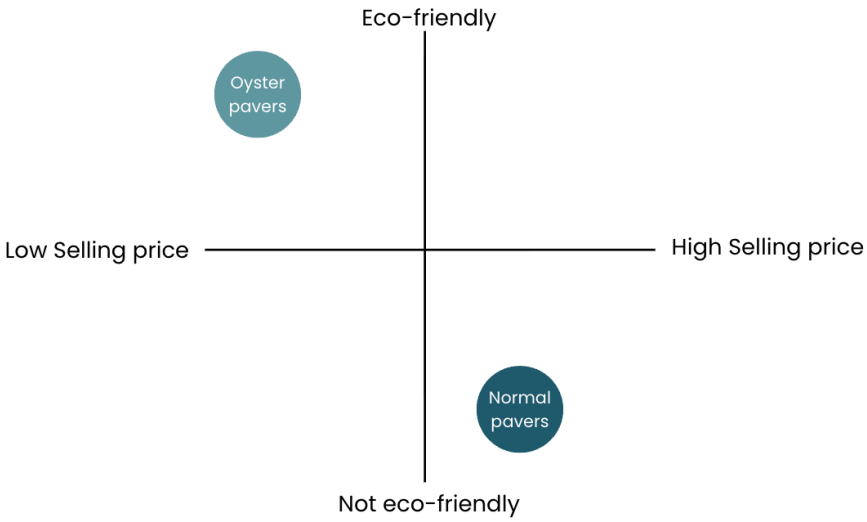
However, the amount of oyster shell waste is directly proportional to the local harvesting rate, which can vary due to factors such as seasonal demand and shifts in the market towards in-shell consumption. (ScienceDirect, 2012)

The following section (Figure **??**) presents a diagram illustrating the SWOT analysis of oyster shell-based pavers and tiles, summarizing the key strengths, weaknesses, opportunities, and threats identified in this study.

Figure 35: SWOT Analysis of Oyster Shell-Based Pavers & Tiles

From the product positioning map (Figure 36), oyster shell-based pavers are positioned in the high eco-friendly and relatively low-price quadrant, indicating a strong value proposition compared to conventional pavers. While normal pavers are perceived as less environmentally friendly and relatively higher in price due to reliance on virgin raw materials and energy-intensive production processes, oyster pavers offer enhanced sustainability through waste recycling and circular economy integration. At the same time, the use of locally sourced oyster shell waste helps reduce raw material and transportation costs, allowing the product to remain price competitive. This positioning highlights oyster shell pavers as a cost-effective and environmentally responsible alternative in the construction materials market.

Figure 36: Example of Product Positioning for Pavers



4.9.2 B2B: Animal feed and Fertilizer

The product involves processing recycled oyster shells into a calcium carbonate supplement in animal feed and as a soil improvement in fertilizer applications, while meeting relevant agricultural and safety standards.

- The pricing strategy focuses on cost competitiveness through locally sourced shell waste, reduced raw material procurement costs, and bulk supply agreements with feed mills, fertilizer producers, and agricultural cooperatives.
- Distribution channels operate through direct B2B channels targeting feed manufacturers, fertilizer companies,nd livestock farms, prioritizing regional markets to ensure consistent supply and lower logistics costs (nd livestock farms, prioritizing regional markets to ensure consistent supply and lower logistics costs (Figure 37).)
- Promotion focuses on nutritional value (calcium supplementation for livestock), soil improvement benefits, cost savings, and alignment with sustainable agriculture and circular economy practices rather than consumer branding.

Figure 37: SWOT Analysis for Agriculture Products

From the product positioning perspective, oyster shell-based agricultural inputs are positioned as highly eco-friendly and cost-competitive alternatives to conventional calcium sources, offering both environmental benefits and economic value for large-scale agricultural operations.

4.9.3 B2C: Coasters and Body Scrub

For the B2C segment, we focused on coasters as a promising product targeting tourists, eco-conscious consumers, and lifestyle buyers. From the 4Ps Analysis, which is a marketing strategic framework used to evaluate products by analyzing four controllable variables, including Product, Price, Place, Promotion, the product consists of water-absorbing coasters made from crushed oyster shells, featuring a natural texture and attractive colors that highlight sustainability and local identity. The pricing strategy is positioned at a mid-range level to reflect handcrafted quality and sustainable materials. Distribution channels include souvenir shops, eco-markets, and online platforms, while promotional strategies focus on sustainability storytelling, waste reduction impact, and local community involvement. The following section (Figure 38) presents a diagram illustrating the SWOT analysis of oyster shell-based coasters, summarizing the key strengths, weaknesses, opportunities, and threats identified in this study.

Figure 38: SWOT Analysis of Oyster Shell-Based Coasters

In terms of product positioning, oyster shell coasters can be positioned as a sustainable, functional, and aesthetically distinctive alternative to traditional ceramic or silicone coasters.

Body scrub was another high-potential B2C product targeting eco-conscious consumers, lifestyle buyers, skincare and wellness users, as well as tourists seeking locally inspired products. The product is developed as an upcycled oyster-shell exfoliating scrub with a moisturizing base. The pricing strategy is positioned at a mid-range level to reflect cosmetic-grade processing and quality control. Distribution channels include eco-beauty stores, spas, community markets, and online platforms, while promotional strategies focus on the waste-reduction aspects, skin benefits, and circular economy impact. The following figure (Figure 39) presents the SWOT analysis of the oyster shell-based body scrub product.

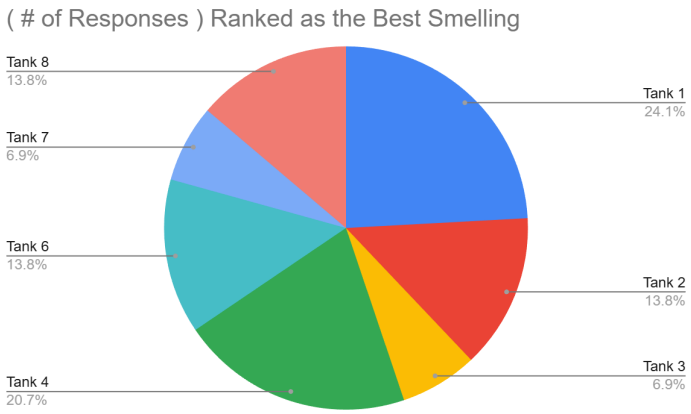
Figure 39: SWOT Analysis of Oyster Shell-Based Body Scrub

In terms of product positioning, the oyster-shell body scrub can be positioned as a sustainable, locally inspired, and performance-driven alternative to conventional exfoliating products.

4.10 Reducing Odor

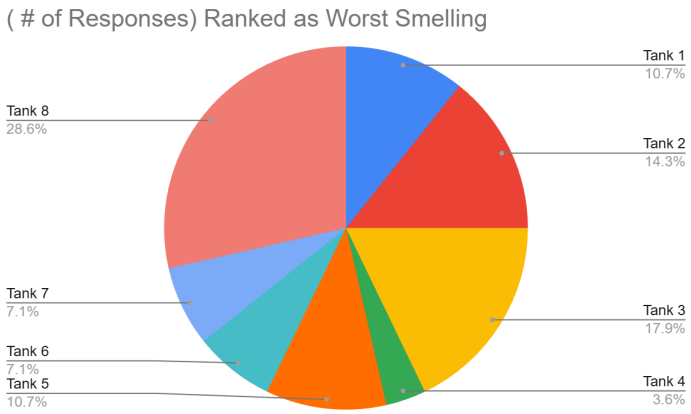
The data below shows the results of the odor testing, with solutions 4, 5, and 6 being identified as the best-smelling solutions (Figure 40). The survey was conducted with 29 participants who were asked to smell each of the different solutions added to the oyster samples.

Figure 40: Percentage of Participants Identifying Each Tank as Best-Smelling



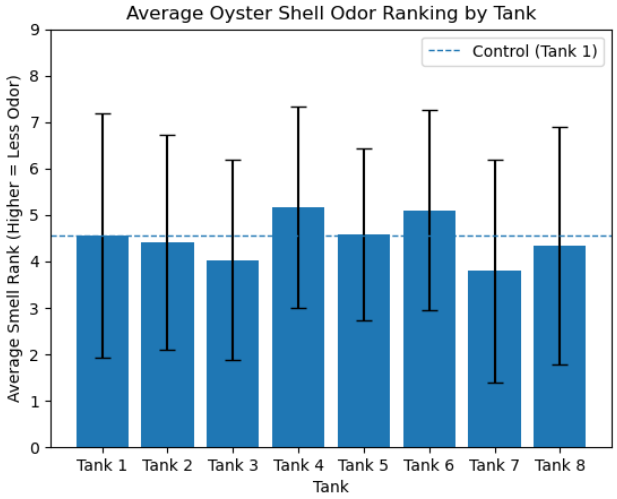
from the project site (Appendix E). From these responses, tank eight was identified as the worst-smelling (Figure **??**). The average score of each sample was averaged to determine which sample would have the broadest appeal (Figure 41).

Figure 41: Percentage of Participants Identifying Each Tank as Worst-Smelling



As seen in Figure 23, most solutions averaged a score near that of the control test of tank 1, 4.55, with tanks 4 and 6 ranking higher and tanks 3 and 8 ranking at or below an average score of 4. As tanks 3 and 8 (9.10

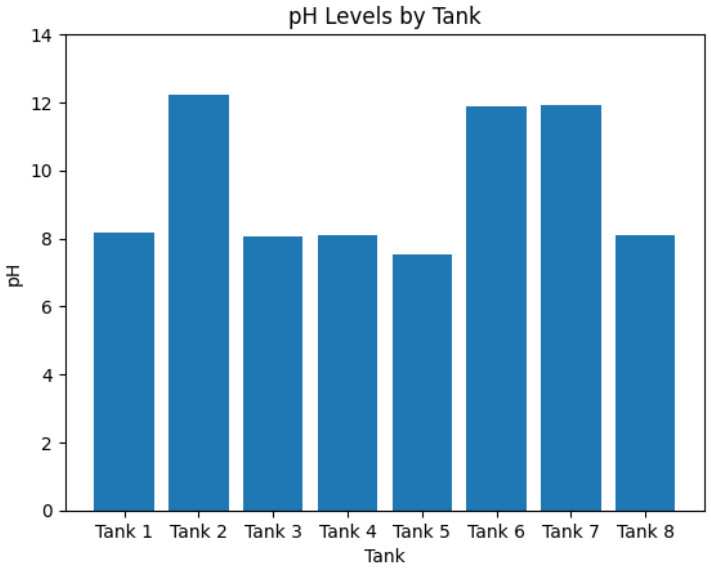
Figure 42: Percentage of Participants Identifying Each Tank as Worst-Smelling



4.11 Environment Testing

To further refine our testing data pH testing of the different solutions was measured to determine which solution would have the least environmental impact. Based on our testing, it can be seen that solution six is the most alkaline, having a pH level similar to soapy water (Figure 43).

Figure 43: pH Levels by Tank



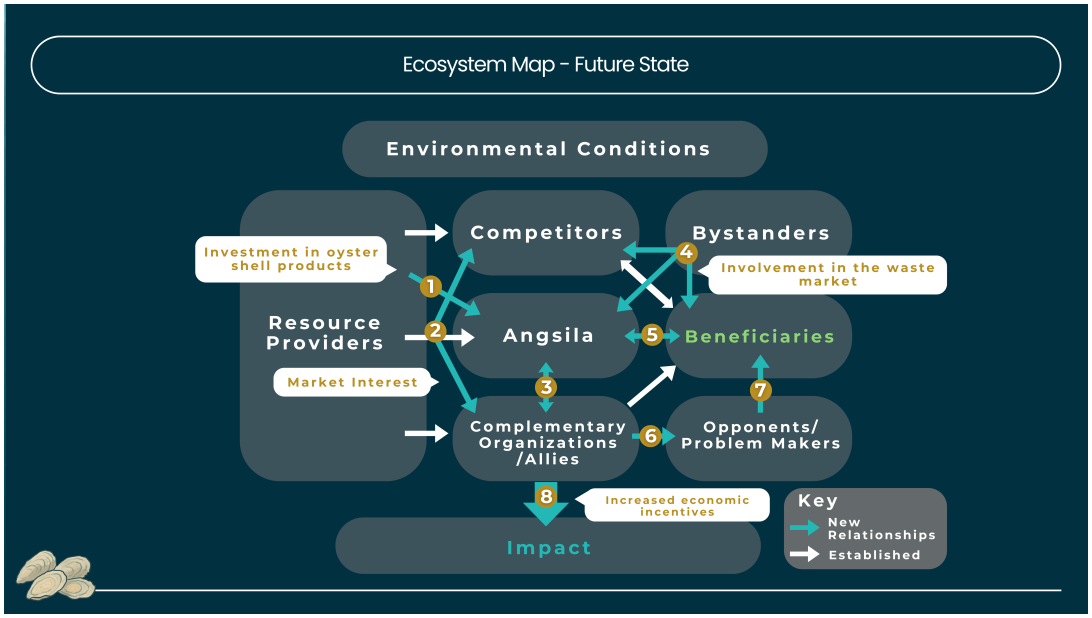
The environmental impact testing, as well as the smell testing, point to solutions four and five being the best candidates for removing odor from oyster shell waste. While this testing does show improved pH and odor levels, the difference between the average score and pH levels is small, requiring further refinement of solutions and further testing.

4.12 Ecosystem Map Future State

The figure below shows the proposed future state, highlighting the anticipated relationships, material flows, and stakeholder interactions after our recommendations are implemented. This ecosystem map highlights the anticipated relationships, new material flows, and stakeholder interaction that would enable a more coordinated and environmentally friendly system for managing oyster shell waste. In the proposed system oyster shells will be cleaned, crushed, and converted into valuable products.

The proposed ecosystem demonstrates how the integration of stakeholders, infrastructure, and market demand can utilize oyster shells as a waste product and transform it into a valuable resource for the community. The full ecosystem map is shown in (Appendix B) with a simplified figure shown below (fig 45).

Figure 44: Ecosystem Map - Future State



5 Conclusions and Recommendations

We framed a structured model that guides the proper collection, cleaning, and storage for further shell processing. In comparison to the current state, the future state represents the potential relationships between stakeholders after implementing our process, highlighting how damaged third parties, competitors, and opponents can turn into beneficiaries and allies.

5.1 Recommendations

In the long term, repurposing oyster shells as raw materials would be the most feasible and scalable approach. The shell powder can be used in the production of value-added products, such as tiles, coasters, and, probably most profitably, pavers, all of which require simple production methods, while demonstrating marketable end products for locals and businesses.

Based on different repurposing methods, such as tiles, pavers, and coasters, different business approaches are needed to ascertain different audiences and markets. For marketing tiles and pavers, it is recommended that businesses have a B2B approach, while coasters have a B2C approach (Pilelienė, 2019). These approaches have different focuses on audiences, which take into consideration factors such as the stakeholders and immediate buyers involved. This will allow businesses to maximize their profits and demands within the market. ref In the near term, to address the initial problem of the shell odor, multiple bacterial treatments such as photosynthetic bacteria inoculum, EM inoculum microorganisms, and calcium hydroxide were evaluated. Our research suggests that photosynthetic bacteria inoculum may be slightly more effective for reducing the shell odor compared to the other mixtures. However, because the rankings were closely distributed and subjective, further research would be needed to determine the overall potential for odor mitigation.

5.2 Limitations

5.2.1 Investment Requirements

Several factors need to be addressed if Angsila is to implement and scale oyster shell repurposing in line with our model. The process requires infrastructure, personnel, and an operating capital budget to properly manage the shell waste. It also requires the development of a multiparty value chain that entails identifying truckers, cleaners, grinders, construction materials manufacturers, marketers, vendors, and end users. It may require finding an entrepreneur or entrepreneurs who can take this on for the municipality.

The first step designated sites where shuckers could collectively dump their waste. However, the absence of infrastructure, such as designated storage and processing facilities, has made it difficult to relocate and manage large volumes of shells. As a result, this has contributed to illegal dumping in areas near the ocean, which further perpetuates oxygen depletion.

There is a shortage of trained and readily available personnel needed to properly manage the oyster shell waste. While responses from our interviews displayed many local vendors and community members' willingness to participate, the lack of a structured system and limited

municipality manpower has made it difficult to guide an organized effort. Without clear coordination and support, these efforts have remained difficult for formal action to take place.

A further limitation of this project is the financial cost associated with the initial process for cleaning and repurposing. Heavy machinery is required for cleaning, grinding, and transporting the shells. The use of this machinery is crucial for improving the scalability of shell management, but the expenses hinder any immediate action. Furthermore, expenses such as maintenance, operation, and additional personnel contribute to financial burdens. This limitation highlights the importance of external funding from stakeholders or partners and the need for a small-scale pilot program. This approach would allow the stakeholders to evaluate the feasibility, cost, and potential before committing to a full-scale expansion.

5.2.2 Lack of Awareness

Along with other limitations, a lack of awareness and urgency for the problem is what continues to create a more uncontrollable issue of shell waste. Based on interviews performed with workers at shucking houses, locals indicated a willingness to improve the shell waste management. However, they had also expressed that they didn't have the knowledge or technical expertise to help mitigate the waste. While the locals now understand there is a waste problem, they feel their knowledge restricts them since this issue is on such a large scale and continues to grow each day. Shucking house workers witness the waste build up first-hand daily, contributing to the overwhelming feeling that this issue has become too large for the locals to handle on their own.

5.2.3 Time Restriction

The restriction of time has been a main focus while planning and completing this project. With seven weeks to carry out our project, there was insufficient time to properly cure the oyster shell-based pavers before conducting the necessary strength tests. Curing is a critical process for ensuring the material achieves the required strength and durability. Without adequate curing time, it was not possible to perform a thorough comparison of the pavers' strength against the standards outlined in the Thai Industrial Standards (TIS). Specifically, TIS 827-2531 sets the compressive strength requirement for interlocking concrete paving blocks at a minimum of 40 MPa. Due to the limited curing period, the tests could not be conducted in full, preventing a proper evaluation of whether the oyster shell-based pavers meet these industry standards.

5.3 Alternatives

5.3.1 Biosurfactants

Another means for cleaning crushed shells is by using biosurfactants, which are currently being researched in the Biology Department at Chulalongkorn University. Biosurfactants are biodegradable and low-toxicity compounds that are used for reducing surface tension. They are made up of microorganisms such as bacteria, yeast, and fungi, and work by weakening ad-

hesives such as oils and organic matter from the surface. For this method, the biosurfactant would help separate the excess oyster meat from the shell, as well as get rid of any residual biofilm (Appendix F).

5.3.2 Shorelines & Breakwaters

Alternatively, oyster shells can be used to construct living shorelines. Many countries have successfully and efficiently implemented these shorelines as it requires a significant amount of shells, minimal cleaning, and are cost-effective. Instead of crushed oyster shells, shoreline barriers utilize whole oyster shells. Durable plastic or biodegradable mesh bags are filled with oyster shells and placed in shallow water. These bags act as a natural wall to absorb the energy of waves (Uptonnes et al., 2018). Their composition of calcium carbonate allows them to stay durable, long after they have been eaten and cleaned. In addition to absorbing energy, the shoreline barriers aid in shoreline erosion and stabilizing sediment. This alternative provides ecological benefits as it can also serve as a habitat for new oysters. Spat, or oyster larvae, attach to the oysters in the mesh bags and can grow on the old ones (Giuliano, 2022). However, by Thai law, oyster shells are classified as solid waste, which prevents utilizing the shells for living shorelines and breakwaters. The Land Excavation and Land Filling Act B.E. 2543 regulates the use of materials, including oyster shells, for land filling purposes at a large scale, which may require notification or approval from municipal authorities and be in compliance with drainage and safety requirements (Appendix G).

5.4 Conclusions

Having the opportunity to work alongside students from Chulalongkorn University in Bangkok, Thailand, allowed us to immerse ourselves in a new culture while strengthening our teamwork and communication skills. Being faced with language barriers presented numerous challenges with miscommunication between our sponsors; however, this encouraged stronger collaboration and adaptability. This experience also exposed us to various perspectives, which broadened our understanding of sustainable waste management and problem-solving approaches. Ultimately, this project not only enhanced our technical and professional skills but also created meaningful experiences that deepened our cultural awareness and appreciation.

References

Ahmad, O. A., Jamal, M. T., Almalki, H. S., Alzahrani, A. H., Alatawi, A. S., & Haque, M. F. (2025). Microplastic pollution in the marine environment: Sources, impacts, and degradation [Accessed: 2025-12-04]. *Journal of Advanced Veterinary and Animal Research*, *12*(1), 260–279. https://doi.org/https://doi.org/10.5455/javar.2025.l893

BCG Thailand. (2021, February). □□□□□□□□□□□□□□□□□□ (circular economy) [Accessed: 2025-12-04]. https://www.bcg.in.th/data-center/articles/circular-economy/

Bellei, P., Torres, I., Solstad, R., & Flores-Colen, I. (2023). Potential use of oyster shell waste in the composition of construction composites: A review [Accessed: 2025-12-04]. *Buildings*, *13*(6), 1546. https://www.mdpi.com/2075-5309/13/6/1546

Chilakala, R., Thannaree, C., Shin, E. J., Thenepalli, T., & Ahn, J.-W. (2019). Sustainable solutions for oyster shell waste recycling in thailand and the philippines [Accessed: 2025-12-04]. *Recycling*, *4*(3), 35. https://doi.org/https://doi.org/10.3390/recycling4030035

Chumkam, S., & Jintasaporn, O. (2020). Effects of pueraria mirifica and calcium supplementation in the diet of spent laying hens on productive performance, egg shell quality, egg internal quality, calcium, and estrogen in blood [Accessed: 2025-12-04]. https://so06.tci-thaijo.org/index.php/vrurdistjournal/article/view/244527/166694

CityPopulation.de. (2025). Thailand: Eastern region (province chonburi) – subdistricts & municipalities (ang sila) [Accessed: 2025-12-04]. https://www.citypopulation.de/en/thailand/eastern/chonburi/2089__ang_sila/

climate-data.org. (2024). Data and graphs for weather & climate in chonburi [Accessed: 2025-12-04]. https://en.climate-data.org/asia/thailand/chon-buri-province/chonburi-4203/

Department of Fisheries (Thailand). (2020). Statistics of marine shellfish culture: Report no. 14/2020 [Accessed: 2025-12-04]. https://gisportal.fisheries.go.th/portal/apps/webappviewer/index.html?id=61050ffe668b408bb9e8e39f4e187a4a

eco cupid. (2024, October). Saensuk city: Mission to zero waste beach [video] [Accessed: 2025-12-04]. https://www.youtube.com/watch?v=bR7dCkUO2xI&t=602s

Explore Thailand Through Photography. (n.d.). Angsila oyster farms (photo collection) [Accessed: 2026-02-24]. https://patricklepetit.jalbum.net/CHONBURI/PHOTOS/MUEANG%20CHONBURI/Angsila/Angsila%20Oyster%20farms/index.html

Food and Agriculture Organization of the United Nations. (2022). The state of world fisheries and aquaculture 2022: Towards blue transformation [Accessed: 2025-12-04]. https://www.fao.org/3/cc0461en/cc0461en.pdf

Giuliano, S. (2022, August). Shell-tering our shores from sea level rise [Accessed: 2025-12-04]. https://www.fws.gov/story/2022-08/shell-tering-our-shores-sea-level-rise

Godoy, C. (2022). Guidelines on the reduction of seafood waste and loss [Accessed: 2025-12-04]. https://www.apec.org/docs/default-source/publications/2022/10/guidelines-on-the-reduction-of-seafood-waste-and-loss/222_ofwg_guidelines-on-the-reduction-of-seafood-waste-and-loss.pdf

IERE. (2024). How do you clean seashells? [Accessed: 2025-12-04]. https://iere.org/how-do-you-clean-seashells

IOM UN Migration. (n.d.). Migration context [Accessed: 2025-12-12]. https://thailand.iom.int/migration-context

Kwon, H.-B., Lee, C.-W., Jun, B.-S., Yun, J., Weon, S.-Y., & Koopman, B. (2004). Recycling waste oyster shells for eutrophication control [Accessed: 2025-12-04]. *Resources, Conservation and Recycling*, *41*(1), 75–82. https://doi.org/https://doi.org/10.1016/j.resconrec.2003.08.005

Li, Y., Sun, J., & Huang, Z. (2021). Conversion of oyster shells into high-value calcium oxide by thermal treatment [Accessed: 2025-12-04]. *Journal of Cleaner Production*. https://www.sciencedirect.com/science/article/abs/pii/S2213343721000919

Ma, W. F., & Li, J. Y. (2022). Analysis of the underlying mechanisms of green tide with a perspective of algae ecophysiology [Accessed: 2025-12-04]. *Ying Yong Sheng Tai Xue Bao = The Journal of Applied Ecology*, *33*(5), 1420–1428. https://doi.org/https://doi.org/10.13287/j.1001-9332.202205.028

Martinez, M. J., Palmer, T. A., Breaux, N. J., & Beseres Pollack, J. (2022). Dynamics of restored and natural oyster reefs after a hurricane [Accessed: 2025-12-04]. *Frontiers in Ecology and Evolution*, *10*, 791739. https://www.frontiersin.org/journals/ecology-and-evolution/articles/10.3389/fevo.2022.791739/full

METALOCUS. (n.d.). Reinventing traditional uses: Angsila oyster scaffolding pavilion by chat architects [Accessed: 2025-12-04]. https://www.metalocus.es/en/news/reinventing-traditional-uses-angsila-oyster-scaffolding-pavilion-chat-architects

Mohan, J., Mohanty, C., Cross, J., & Khajuria, A. (2023). Mapping 3r and circular economy policy implementation in asia and the pacific [Accessed: 2025-12-04]. *Circular Economy and Sustainability*. https://www.researchgate.net/publication/373831879_Mapping_3R_and_Circular_Economy_Policy_Implementation_in_Asia_and_the_Pacific

Morris, J. P., Backeljau, T., & Chapelle, G. (2023). Shells from aquaculture: A valuable biomaterial, not a nuisance waste product [Accessed: 2025-12-04]. *Blue Foods*, *2*, 100071. https://www.sciencedirect.com/science/article/pii/S2667010023000719

NANOTEC. (2025, July). Nanotec nstda and partners transform shell waste into fire-resistant paint for safer schools [Accessed: 2025-12-04]. https://www.nanotec.or.th/en/nanotec-nstda-and-partners-transform-shell-waste-into-fire-resistant-paint-for-safer-schools/

NASA. (2023). Technology readiness levels [Accessed: 2025-12-04]. https://www.nasa.gov/directorates/somd/space-communications-navigation-program/technology-readiness-levels

National Audubon Society. (2024, September). Five things we learned in the aftermath of hurricanes harvey and irma [Accessed: 2025-12-04]. https://www.audubon.org/news/five-things-we-learned-aftermath-hurricanes-harvey-and-irma

Pileliene, L. (2019). Principal differences between b2b and b2c marketing communication processes [Accessed: 2025-12-04]. *Management of Organizations: Systematic Research*. https://doi.org/https://doi.org/10.1515/MOSR-2019-0005

Rodríguez-Navarro, A., et al. (2002). Chemical–mechanical characteristics of crushed oyster shell [Accessed: 2025-12-04]. *Waste Management*. https://www.sciencedirect.com/science/article/abs/pii/S0956053X0200159

Ruslan, H., Muthusamy, K., Mohsin, S., Jose, R., & Omar, R. (2022). Oyster shell waste as a concrete ingredient: A review [Accessed: 2025-12-04]. https://www.sciencedirect.com/science/article/pii/S2214785321012906

Sampantamit, T., Ho, L., Lachat, C., Sutummawong, N., Sorgeloos, P., & Goethals, P. (2020). Aquaculture production and its environmental sustainability in thailand: Challenges and potential solutions [Accessed: 2025-12-04]. *Sustainability*, *12*(5), 2010. https://doi.org/https://doi.org/10.3390/su12052010

Seo, J. H., Park, S. M., Yang, B. J., & Jang, J. G. (2019). Calcined oyster shell powder as an expansive additive in cement mortar [Accessed: 2025-12-04]. *Materials*, *12*(8), 1322. https://doi.org/10.3390/ma12081322

Siddiqua, A., Hahladakis, J. N., & Al-Attiya, W. A. K. A. (2022). An overview of the environmental pollution and health effects associated with waste landfilling and open dumping [Accessed: 2025-12-04]. *Environmental Science and Pollution Research*. https://pmc.ncbi.nlm.nih.gov/articles/PMC9399006/

Szuster, B., & Flaherty, M. (2005). Women oyster vendors in eastern thailand [Accessed: 2025-12-04]. https://library.enaca.org/AquacultureAsia/Articles/Jan-March-2005/Jan-March-05.pdf

Thailand Government. (1992a). Enhancement and conservation of the national environmental quality act b.e. 2535 (1992) [Accessed: 2025-12-04]. https://www.aerothai.co.th/sites/default/files/files/related-laws/%E0%B8%9E%E0%B8%A3%E0%B8%B0%E0%B8%A3%E0%B8%B2%E0%B8%8A%E0%B8%9A%E0%B8%B1%E0%B8%8D%E0%B8%8D%E0%B8%B1%E0%B8%95%E0%B8%B4%E0%B8%AA%E0%B9%88%E0%B8%87%E0%B9%80%E0%B8%AA%E0%B8%A3%E0%B8%B4%E0%B8%A1%E0%B9%81%E0%B8%A5%E0%B8%B0%E0%B8%A3%E0%B8%B1%E0%B8%81%E0%B8%A9%E0%B8%B2%E0%B8%84%E0%B8%B8%E0%B8%93%E0%B8%A0%E0%B8%B2%E0%B8%9E%E0%B8%AA%E0%B8%B4%E0%B9%88%E0%B8%87%E0%B9%81%E0%B8%A7%E0%B8%94%E0%B8%A5%E0%B9%89%E0%B8%AD%E0%B8%A1%20%E0%B8%9E.%E0%B8%A8.2535.pdf

Thailand Government. (1992b). Public health act b.e. 2535 (1992) [Accessed: 2025-12-04]. https://epa.onep.go.th/storage/upload/law/%E0%B8%9E%E0%B8%A3%E0%B8%B0%E0%B8%A3%E0%B8%B2%E0%B8%8A%E0%B8%9A%E0%B8%B1%E0%B8%8D%E0%B8%8D%E0%B8%B1%E0%B8%95%E0%B8%B4%E0%B8%81%E0%B8%B2%E0%B8%A3%E0%B8%AA%E0%B8%B2%E0%B8%98%E0%B8%B2%E0%B8%A3%E0%B8%93%E0%B8%AA%E0%B8%B8%E0%B8%82%20%E0%B8%9E.%E0%B8%A8.2535.pdf

Thailand Government. (2000). Act on excavation and land filling b.e. 2543 (2000) [Accessed: 2025-12-04]. https://www.tuda.or.th/wp-content/uploads/2018/04/%E0%B8%9E%

E0%B8%A3%E0%B8%9A.%E0%B8%81%E0%B8%B2%E0%B8%A3%E0%B8%82%E0%B8%B8%E0%B8%94%E0%B8%94%E0%B8%B4%E0%B8%99%E0%B9%81%E0%B8%A5%E0%B8%B0%E0%B8%96%E0%B8%A1%E0%B8%94%E0%B8%B4%E0%B8%99%E0%B8%9E.%E0%B8%A8.2543.pdf

Tourism Authority of Thailand. (n.d.). Ang sila (attraction information) [Accessed: 2025-12-04]. https://www.tourismthailand.org/Attraction/ang-sila

United Nations. (2024). Sustainable development goal 12: Responsible consumption and production [Accessed: 2025-12-04]. https://sdgs.un.org/goals/goal12

United Nations Children's Fund & College of Population Studies, Social Research Institute, Chula Unisearch. (2023). In-depth research on youth not in employment, education or training (neet) in thailand [Accessed: 2025-12-04]. https://www.unicef.org/thailand/media/10746/file/In-depth%20research%20on%20youth%20NEET%20in%20Thailand.pdf

Upton, J., Weber, S., & Kargbo, C. (2018, July). "living shorelines" use oyster shells and marsh grass to reverse coastal erosion [Accessed: 2025-12-04]. https://www.pbs.org/newshour/show/living-shorelines-use-oyster-shells-and-marsh-grass-to-reverse-coastal-erosion

U.S. Department of Health & Human Services, Office for Human Research Protections. (1979). Ethical principles and guidelines for the protection of human subjects of research (the belmont report) [Updated page accessed: 2025-12-04]. https://www.hhs.gov/ohrp/regulations-and-policy/belmont-report/read-the-belmont-report/index.html#xbasic

U.S. Environmental Protection Agency. (2017). Nutrient pollution: The problem [Accessed: 2025-12-04]. https://www.epa.gov/nutrientpollution/problem

Wisdom Library. (2025, November). Ban angsila (city information) [Accessed: 2025-12-04]. https://www.wisdomlib.org/cities/ban-ang-sila-16811

Woo, H.-E., Patil, M. P., Kim, Y.-R., Joo, Y., Kim, J.-O., & Kim, K. (2026). Application of oyster shells for environmental remediation of sewage-affected coastal sediments in masan bay, korea [Accessed: 2025-12-04]. *Marine Pollution Bulletin*, *222*(Part 1), 118671. https://doi.org/https://doi.org/10.1016/j.marpolbul.2025.118671

World Health Organization. (2003). Hydrogen sulfide: Human health aspects (concise international chemical assessment document 53) [Accessed: 2025-12-04]. https://iris.who.int/server/api/core/bitstreams/ab85a2cd-2221-4dba-9895-39ff6cbe79ed/content

Yoshioka, N., Era, M., & Sasaki, D. (2021). Towards integration of climate disaster risk and waste management: A case study of urban and rural coastal communities in the philippines [Accessed: 2025-12-04]. *Sustainability*, *13*(4), 1624. https://doi.org/https://doi.org/10.3390/su13041624

Zhang, G., & Zhang, J. (2018). Nano calcium carbonate from golden apple snail and oyster shells as reinforcing filler for PVC [Accessed: 2025-12-04]. https://digital.car.chula.ac.th/chulaetd/61183/

Appendices

A Related Laws and Regulations

A.1 Public Health Act B.E. 2535 (1992)

Under the Public Health Act B.E. 2535 (1992) of Thailand, which governs waste management and sanitation at the local level. Section 18 of the Act assigns responsibility for the collection, transportation, and disposal of waste within a local administrative area to the relevant local authority. Local governments may carry out these duties independently, in cooperation with other government agencies, or through formal agreements with neighboring authorities. The Act also permits local authorities to delegate waste management activities to private operators under official supervision, or to grant licenses in accordance with Section 19.

Section 19 states that any person operating a waste collection, transportation, or disposal business for compensation must obtain a license from the local authority. This requirement ensures that waste management activities comply with public health and sanitation standards and are subject to regulatory oversight.

Furthermore, Section 20 empowers local authorities to issue local regulations to maintain cleanliness and order in waste management practices. This includes the authority to prohibit unauthorized dumping in public areas, designate specific disposal sites, require appropriate waste containers, establish standards for waste handling, and set service fees within ministerial limits. Local authorities may also impose additional measures necessary to safeguard public health and environmental sanitation.

A.2 Land Excavation and Land Filling Act, B.E. 2543 (2000)

Under the Land Excavation and Land Filling Act B.E. 2543, the use of oyster shells to raise ground levels falls within the legal definition of "land filling" as the Act covers the use of land or other materials to elevate ground height. Therefore, oyster shells qualify as fill material, and landowners must comply with applicable notification and permitting requirements. Claims that shell-based filling does not constitute land filling due to the absence of soil are not legally valid.

Section 26 requires notification to the local authority when land filling exceeds specified height and area thresholds (generally over 2,000 square meters). Large-scale shell filling without proper notification may constitute a procedural violation.

Section 29 mandates adequate drainage systems to prevent damage to adjacent land. Improper shell filling that disrupts natural water flow, causes flooding, or creates a nuisance may result in legal liability.

Section 30 further requires safety and nuisance prevention measures. Offensive odors arising from decomposing residual organic matter or dust generated from broken shells clearly constitute a nuisance under this section. In such cases, authorities may order suspension and corrective action to protect public health and safety.

A.3 Enhancement and Conservation of the National Environmental Quality Act B.E. 2535 (1992)

Under the Enhancement and Conservation of National Environmental Quality Act B.E. 2535, the landfilling of oyster shells may fall within the legal definitions of "pollution" and deterioration of "environmental quality" if such activities result in adverse environmental impacts. Oyster shells containing residual organic matter may decompose and generate pollutants, leachate, or elevated biochemical oxygen demand (BOD), potentially contaminating soil, surface water, or groundwater. Pursuant to Sections 54 and 55, pollution sources must comply with prescribed environmental standards and implement appropriate treatment or control measures to prevent pollutant discharge beyond legal limits. Accordingly, improperly managed shell-based landfilling may constitute a violation if it causes environmental degradation, offensive odors, or water quality impacts exceeding regulatory standards.

B Ecosystem Map

B.1 Ecosystem Map Current State

The figure below presents the complete current state ecosystem map. This image provides additional detail about each stakeholder group and explains the environmental conditions and impacts associated with the existing system. The arrows illustrate how each group interacts within the ecosystem. The directionality of the arrows indicates whether these interactions occur in a single direction or represent mutual exchanges in which both groups influence one another. By creating a visual representation of the current system, the ecosystem map is able to inform stakeholders and bystanders of their current position in the problem. In the current state, resource providers possess the capacity to offer financial support, personnel, and machinery to help coordinate a process for repurposing shells. However, the competitors and allies dependent on these resources lack the connections needed to advance towards a solution. As a result, the system continues to perpetuate negative externalities towards damaged third parties such as Angsila residents, nearby households, shucking workers, and ecosystems. The connection between these groups and damaged third parties remains because there is no present system that encourages collaboration or sustainable practices, and instead resists formal regulation. Furthermore, opponents such as informal waste disposers work independently and remain disconnected from available resources. This isolation enables informal dumping practices to persist without accountability, directly contributing to the environmental harm experienced by these third parties. The absence of these linkages highlights the gaps between organizations that are contributing to the problem, yet are the least integrated in addressing the problem

Figure 45: Ecosystem Map - Current State

B.2 Ecosysem Map Future State

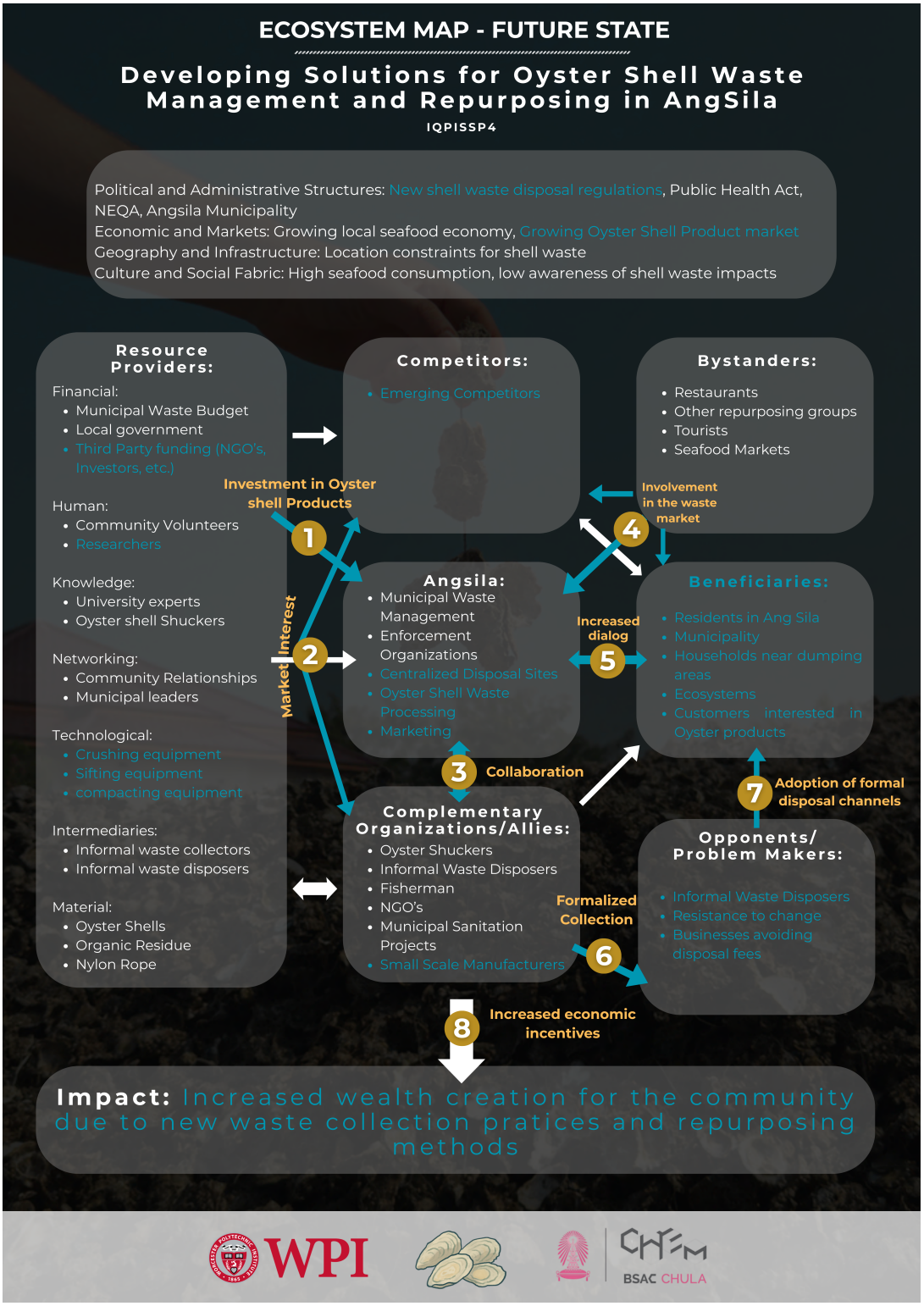
The figure below represents the full future state ecosystem map, which highlights the changes from the current state, which creates a negative impact, to a state that increases opportunities for those involved in the project. As can be seen, new arrows have been added, and new directions of interactions have developed.

- Arrow one represents the investment of financial resources into the municipality to sup-

port the development of oyster shell repurposing methods.

- Arrow two reflects the growing market interest in environmentally sustainable products, which can be utilized to develop oyster shell products that encourage consumer demand.
- Arrow three demonstrates the increased collaboration between the municipality and complementary organizations that will be established to sustain a productive and scalable industry.
- Arrow four illustrates the involvement of bystander groups who become engaged due to the economic and environmental potential of the new repurposing system.
- Arrow five highlights enhanced dialogue among the municipality, residents, and tourists, fostering trust and encouraging reinvestment from groups that were previously negatively affected.
- Arrow six shows how the implementation of a formalized shell collection procedure transforms former opponents or contributors to the problem into active participants in the repurposing program.
- Arrow seven emphasizes how previously harmful actors are incentivized to contribute positively, directly benefiting groups that were once disadvantaged.
- Arrow eight illustrates how the system changes, and our recommendations collectively produce a desirable, community-wide impact that supports long-term environmental, social, and economic sustainability.

Figure 46: Ecosystem Map - Future State



C Technology Readiness Level (TRL)

- **TRL 1: Basic Observations**
  Foundational research is conducted on the composition and structurability of oyster shells to support future research and development.

- **TRL 2: Concept Formulation**
  Based on preliminary research, formulations for repurposing oyster shells into coasters, tiles, and pavers are developed, taking binding agents and processing methods into consideration.

- **TRL 3: Proof of Concept**
  Prototypes are produced using the proposed formulations to identify defects and assess overall feasibility.

- **TRL 4: Laboratory Validation**
  Material and structural properties of the products are tested to ensure performance requirements are met.

- **TRL 5: Environmental Validation**
  Products such as coasters are tested in controlled, real-world environments (e.g., cafés) to evaluate durability and functionality. User feedback is collected to identify modifications required for daily-use standards.

- **TRL 6: Model Demonstration**
  A small-scale production model is developed to demonstrate the full manufacturing process. This stage evaluates time efficiency, resource requirements, and workflow optimization.

- **TRL 7: Pilot Program**
  Incorporating feedback from TRL 5, upgraded prototypes are deployed through a pilot program and integrated among users.

- **TRL 8: Product Certification**
  Small-scale production is refined and finalized to meet applicable quality, safety, and manufacturing standards.

- **TRL 9: Product Launch**
  Full-scale production is launched, accounting for target audiences, branding, and market considerations.

D Interview Questions

D.1 D1: Questions for Angsila Town Muncipality Officers

1. Can you provide an overview of how oyster shells are currently managed in Angsila?
   (a) Which areas generate the most oyster shell waste?
   (b) What records (official or unofficial) of quantities collected or disposed of do you have?
2. What are the specific policies, regulations, or guidelines regarding oyster shell waste in Angsila?
   (a) Which department is responsible for monitoring or enforcing these rules?
3. How are oyster shells usually collected and transported from shucking areas or markets?
   (a) Does the municipality provide any collection service or facilities specifically for shells?
4. Where are oyster shells currently disposed of or stored?
   (a) Are there designated dumping sites or temporary storage areas for shells?
   (b) What environmental or health problems have been observed at these locations?
5. What are the main challenges or issues the municipality faces in managing oyster shell waste?
6. What are the existing initiatives or past attempts to recycle or reuse oyster shells in Angsila?
   (a) What were the outcomes or lessons learned?
7. Does the municipality collaborate with universities, private companies, or other organizations on oyster shell management or product development?
8. How do you perceive the attitudes of residents and oyster-related businesses toward shell waste management?
9. In your opinion, what would be the most important elements of an effective oyster shell waste management system for Angsila?
10. What kind of support would the municipality need to implement a more sustainable system for oyster shell waste?

D.2 D2 : Questions for Oyster Farmers and Shuckers/Vendors

1. Can you describe your oyster farming or shucking activities?
   (a) Approximately how many oysters do you harvest or shuck per day in peak season and low season?
2. What do you currently do with oyster shells after removing the meat?
   (a) a. Where are the shells usually dumped or stored?
   (b) b. Who decides on shell disposal?
3. Have you experienced any problems related to oyster shells, such as smell, insects, or lack of space?
   (a) How do these problems affect your work or your neighbours?
4. Have you ever sold or given oyster shells to someone for reuse (e.g., shrimp farms, feed, construction, handicrafts)?
   (a) If yes, how often does this happen and at what price?
   (b) If not, what are the reasons?
5. Are there any costs associated with managing or disposing of oyster shells for your household/business (e.g., labour, transportation, cleaning)?
6. Have you received any information, training,g or support from the municipality or other organizations about oyster shell waste management?
7. If a new system were introduced to collect and recycle oyster shells. What conditions would make you willing to participate?
8. Do you think oyster shells could become a source of extra income for farmers or shuckers in Angsila?
   (a) What types of products or uses would you be most interested in?
9. From your perspective, what is the most urgent problem that needs to be solved regarding oyster shell waste in Angsila?

D.3 D3 : Questions for Residents Living Near Shell Piles or Shucking Areas

1. What kind of odor-related problems have you experienced as a result of oyster shell waste?
2. How do these problems affect your daily life, health, or business activities?

3. Have you ever raised complaints or concerns about oyster shell waste to local authorities or community leaders?

    (a) a. If yes, what responses or actions did you receive?

    (b) b. If no, why not?

4. Are you aware of any attempts by the municipality or community to improve oyster shell waste management in your area?

5. In your opinion, who should be mainly responsible for managing oyster shell waste in Angsila?

6. Would you be willing to participate in community activities or meetings to improve oyster shell waste management?

    (a) a. What kind of support or information would you need?

7. What changes or solutions would you like to see to reduce the negative impacts of oyster shell waste in your community?

E Impact Tests

Odor Survey

Survey provided to participants in a smell testing survey. Volunteer participants were utilized with no financial or material incentive. Buckets were placed in numerical order one through eight. Participants smelled each bucket in order and ranked each bucket with a score of one through eight. To avoid ties, the survey required unique scores for each bucket.

Table of Responses to Smell Survey

The table below shows the results of the survey above, with a total of 29 responses to the survey testing. The results are displayed to provide quantitative data on the results of the testing. All responses are anonymous to those conducting the survey and record no personal information. Finally, all scores from all 29 responses were averaged in order to decrease personal biases.

Table 5: Stakeholder ranking responses.

Timestamp	R1	R2	R3	R4	R5	R6	R7	R8	Avg
12/2/2026 9:38:24	3	6	8	7	2	5	4	1	4.55
12/2/2026 9:39:16	3	4	1	8	6	2	7	5	4.41
12/2/2026 9:39:43	3	4	6	7	5	1	8	2	4.03
13/2/2026 14:52:11	6	2	1	8	4	5	3	7	5.17
13/2/2026 14:54:16	1	2	3	4	7	8	5	6	4.59
13/2/2026 14:54:39	7	3	4	8	5	6	2	1	5.10
13/2/2026 14:56:36	5	1	3	8	7	6	4	2	4.21
13/2/2026 14:57:45	7	8	6	4	5	3	2	1	3.93
13/2/2026 14:59:30	3	1	2	6	4	5	7	8	4.55
13/2/2026 14:59:35	8	7	6	5	3	1	2	4	4.41
13/2/2026 15:00:08	8	5	6	7	4	3	2	1	4.03
13/2/2026 15:00:47	6	7	4	3	1	5	2	8	5.17
13/2/2026 15:03:09	1	3	2	4	6	8	5	7	4.59
13/2/2026 15:03:32	8	4	2	1	3	7	6	5	5.10
13/2/2026 15:03:33	8	7	6	5	4	3	2	1	4.21
13/2/2026 15:04:54	5	4	1	6	7	8	3	2	3.93
13/2/2026 15:05:11	8	4	6	3	5	7	1	2	4.55
13/2/2026 15:05:34	6	8	3	2	7	5	4	1	4.41
13/2/2026 15:09:46	8	3	4	6	7	2	1	5	4.03
13/2/2026 15:11:41	2	1	3	5	4	6	7	8	5.17
13/2/2026 15:13:36	2	8	7	6	5	3	4	1	4.59
13/2/2026 15:17:11	4	5	1	3	6	7	8	2	5.10
17/2/2026 10:09:47	1	3	5	2	6	7	4	8	4.21
17/2/2026 10:10:01	8	6	4	2	1	3	5	7	3.93
17/2/2026 10:10:19	3	1	8	5	4	6	7	2	4.55
17/2/2026 10:10:35	1	6	5	8	3	7	4	2	4.41
17/2/2026 10:10:57	2	3	1	6	5	8	4	7	4.03
17/2/2026 10:11:11	3	4	5	8	1	6	2	7	5.17
17/2/2026 10:11:37	2	8	4	3	6	5	7	1	4.59

pH Testing Results

The figure below shows the results of the pH testing conducted on solutions one through eight. One round of pH testing was conducted to ascertain the environmental impact different solutions would have.

F Oyster Shell Coaster Testing — TRL 5

Oyster shell coasters were tested at Power of Cups Cafe to evaluate their performance in a real-world use environment, corresponding to Technology Readiness Level (TRL) 5, which focuses on validation in relevant operational settings. The coasters were incorporated into daily café operations and used with both hot and cold beverages under normal service conditions.

Café workers reported that the coasters functioned effectively as beverage coasters and were suitable for routine use. Staff noted that they liked that the oyster shell coasters, unlike cork coasters, did not require daily washing, reducing routine maintenance and cleaning effort.

Posters were displayed in the café to raise awareness about oyster shell waste repurposing, and a short project video associated with the deployment received approximately 1,700 views in 6 days. The café setting allowed the coasters to serve both a functional role and an awareness-raising role by prompting conversations about sustainable material reuse.

Although testing was conducted in Bangkok, this could be applied to Ang Sila, where cafés and small food-service businesses are common. The successful performance of the coasters in an active café environment demonstrated their potential as a low-barrier product for both functional use and public awareness, supporting their classification at TRL 5.

G Biosurfactants

Biosurfactants are active and low-toxicity compounds made of microorganisms such as bacteria and yeast, that help reduce surface tension between two liquids, or a liquid and a solid. In order to produce biosurfactants, the microorganisms are cultivated in renewable sources such as carbon and nitrogen, and further fermented. This process can only be carried out under certain temperatures, pH levels, and oxygen levels (Sarubbo et al., 2022). The molecular structure of biosurfactants consists of both hydrophilic and hydrophobic regions, allowing them to emulsify oils and loosen organic residue. This makes them particularly effective at breaking down thin layers of biofilms. In addition, biosurfactants contain antimicrobial properties that assist in preventing bacterial growth.

These surfactants have various applications in industries such as petroleum, food, agricultural, waste, and cosmetics. In the beauty industry, biosurfactants are used in cleansers, body washes, and shampoos to help remove oil and dirt (Sarubbo et al., 2022). In the context of oyster shell waste management, biosurfactants present a potential method for improving the efficiency of the cleaning process. Their ability to break down biofilm can be applicable for removing debris such as the excess oyster shell meat, while also reducing odor caused by microbial decomposition. This would enhance sanitation prior to the repurposing process of the shells. Additionally, their established use in cosmetics supports the safety procedures of using oyster shell powder in beauty products such as exfoliants.

Overall, the integration of biosurfactants into the shell-cleaning process offers a sustainable and feasible method that expands repurposing opportunities, especially those that require a higher standard of sanitation than our traditional cleaning methods provide.